

Example 2 Cramer's Rule is inefficient for numbers but it is well suited to letters. For $n = 2$, find the columns of A^{-1} by solving $AA^{-1} = I$:

$$\text{Columns of } I \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Those share the same A . We need five determinants for x_1, x_2, y_1, y_2 :

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \text{ and } \begin{vmatrix} 1 & b \\ 0 & d \end{vmatrix} \quad \begin{vmatrix} a & 1 \\ c & 0 \end{vmatrix} \quad \begin{vmatrix} 0 & b \\ 1 & d \end{vmatrix} \quad \begin{vmatrix} a & 0 \\ c & 1 \end{vmatrix}$$

The last four are $d, -c, -b$, and a . (They are the cofactors!) Here is A^{-1} :

$$x_1 = \frac{d}{|A|}, \quad x_2 = \frac{-c}{|A|}, \quad y_1 = \frac{-b}{|A|}, \quad y_2 = \frac{a}{|A|}, \text{ and then } A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

I chose 2 by 2 so that the main points could come through clearly. The new idea is the appearance of the cofactors. When the right side is a column of the identity matrix I , the determinant of each matrix B_j in Cramer's Rule is a cofactor.

You can see those cofactors for $n = 3$. Solve $AA^{-1} = I$ (first column only):

$$\begin{array}{l} \text{Determinants} \\ = \text{Cofactors of } A \end{array} \quad \begin{vmatrix} 1 & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & a_{32} & a_{33} \end{vmatrix} \quad \begin{vmatrix} a_{11} & 1 & a_{13} \\ a_{21} & 0 & a_{23} \\ a_{31} & 0 & a_{33} \end{vmatrix} \quad \begin{vmatrix} a_{11} & a_{12} & 1 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & 0 \end{vmatrix} \quad (5)$$

That first determinant $|B_1|$ is the cofactor C_{11} . The second determinant $|B_2|$ is the cofactor C_{12} . Notice that the correct minus sign appears in $-(a_{21}a_{33} - a_{23}a_{31})$. This cofactor C_{12} goes into the 2, 1 entry of A^{-1} —the first column! So we transpose the cofactor matrix, and as always we divide by $\det A$.

The i, j entry of A^{-1} is the cofactor C_{ji} (not C_{ij}) divided by $\det A$:

$$\text{FORMULA FOR } A^{-1} \quad (A^{-1})_{ij} = \frac{C_{ji}}{\det A} \quad \text{and} \quad A^{-1} = \frac{C^T}{\det A}. \quad (6)$$

The cofactors C_{ij} go into the "cofactor matrix" C . Its transpose leads to A^{-1} . To compute the i, j entry of A^{-1} , cross out row j and column i of A . Multiply the determinant by $(-1)^{i+j}$ to get the cofactor, and divide by $\det A$.

Check this rule for the 3, 1 entry of A^{-1} . This is in column 1 so we solve $Ax = (1, 0, 0)$. The third component x_3 needs the third determinant in equation (5), divided by $\det A$. That third determinant is exactly the cofactor $C_{13} = a_{21}a_{32} - a_{22}a_{31}$. So $(A^{-1})_{31} = C_{13}/\det A$ (2 by 2 determinant divided by 3 by 3).

Summary In solving $AA^{-1} = I$, the columns of I lead to the columns of A^{-1} . Then Cramer's Rule using $b =$ columns of I gives the short formula (6) for A^{-1} .

Direct proof of the formula $A^{-1} = C^T / \det A$ The idea is to multiply A times C^T :

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} = \begin{bmatrix} \det A & 0 & 0 \\ 0 & \det A & 0 \\ 0 & 0 & \det A \end{bmatrix}. \quad (7)$$

Row 1 of A times column 1 of the cofactors yields the first $\det A$ on the right:

$$a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} = \det A \quad \text{by the cofactor rule.}$$

Similarly row 2 of A times column 2 of C^T (transpose) yields $\det A$. The entries a_{2j} are multiplying cofactors C_{2j} as they should, to give the determinant.

How to explain the zeros off the main diagonal in equation (7)? Rows of A are multiplying cofactors from *different* rows. Why is the answer zero?

Row 2 of A

Row 1 of C

$$a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13} = 0. \quad (8)$$

Answer: This is the cofactor rule for a new matrix, when the second row of A is copied into its first row. The new matrix A^* has two equal rows, so $\det A^* = 0$ in equation (8). Notice that A^* has the same cofactors C_{11}, C_{12}, C_{13} as A —because all rows agree after the first row. Thus the remarkable multiplication (7) is correct:

$$AC^T = (\det A)I \quad \text{or} \quad A^{-1} = \frac{C^T}{\det A}.$$

Example 3 The “sum matrix” A has determinant 1. Then A^{-1} contains cofactors:

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \text{has inverse} \quad A^{-1} = \frac{C^T}{1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

Cross out row 1 and column 1 of A to see the 3 by 3 cofactor $C_{11} = 1$. Now cross out row 1 and column 2 for C_{12} . The 3 by 3 submatrix is still triangular with determinant 1. But the cofactor C_{12} is -1 because of the sign $(-1)^{1+2}$. This number -1 goes into the $(2, 1)$ entry of A^{-1} —don't forget to transpose C .

The inverse of a triangular matrix is triangular. Cofactors give a reason why.

Example 4 If all cofactors are nonzero, is A sure to be invertible? *No way.*

Area of a Triangle

Everybody knows the area of a rectangle—base times height. The area of a triangle is *half* the base times the height. But here is a question that those formulas don't answer. *If we know the corners (x_1, y_1) and (x_2, y_2) and (x_3, y_3) of a triangle, what is the area?* Using the corners to find the base and height is not a good way.

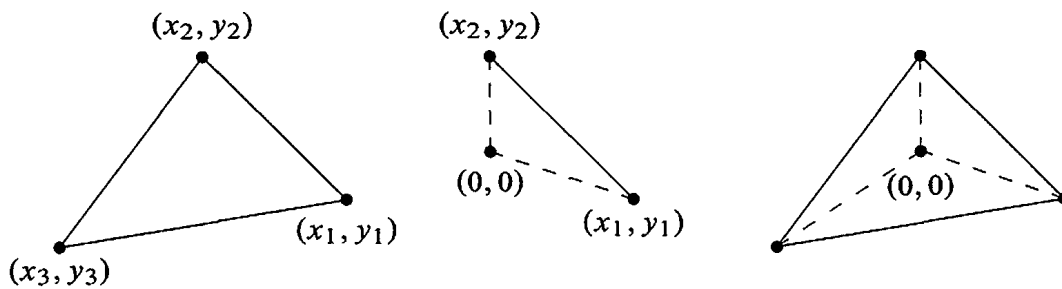


Figure 5.1: General triangle; special triangle from $(0, 0)$; general from three specials.

Determinants are much better. The square roots in the base and height cancel out in the good formula. **The area of a triangle is half of a 3 by 3 determinant.** If one corner is at the origin, say $(x_3, y_3) = (0, 0)$, the determinant is only 2 by 2.

The triangle with corners (x_1, y_1) and (x_2, y_2) and (x_3, y_3) has area = $\frac{\text{determinant}}{2}$:

$$\text{Area of triangle} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \quad \text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} \quad \text{when } (x_3, y_3) = (0, 0).$$

When you set $x_3 = y_3 = 0$ in the 3 by 3 determinant, you get the 2 by 2 determinant. These formulas have no square roots—they are reasonable to memorize. The 3 by 3 determinant breaks into a sum of three 2 by 2's, just as the third triangle in Figure 5.1 breaks into three special triangles from $(0, 0)$:

$$\begin{array}{l} \text{Cofactors of} \\ \text{column 3} \end{array} \quad \text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \begin{array}{l} +\frac{1}{2}(x_1y_2 - x_2y_1) \\ +\frac{1}{2}(x_2y_3 - x_3y_2) \\ +\frac{1}{2}(x_3y_1 - x_1y_3). \end{array} \quad (9)$$

If $(0, 0)$ is outside the triangle, two of the special areas can be negative—but the sum is still correct. The real problem is to explain the special area $\frac{1}{2}(x_1y_2 - x_2y_1)$.

Why is this the area of a triangle? We can remove the factor $\frac{1}{2}$ and change to a parallelogram (twice as big, because the parallelogram contains two equal triangles). We now prove that the parallelogram area is the determinant $x_1y_2 - x_2y_1$. This area in Figure 5.2 is 11, and therefore the triangle has area $\frac{11}{2}$.

Proof that a parallelogram starting from $(0, 0)$ has area = 2 by 2 determinant.

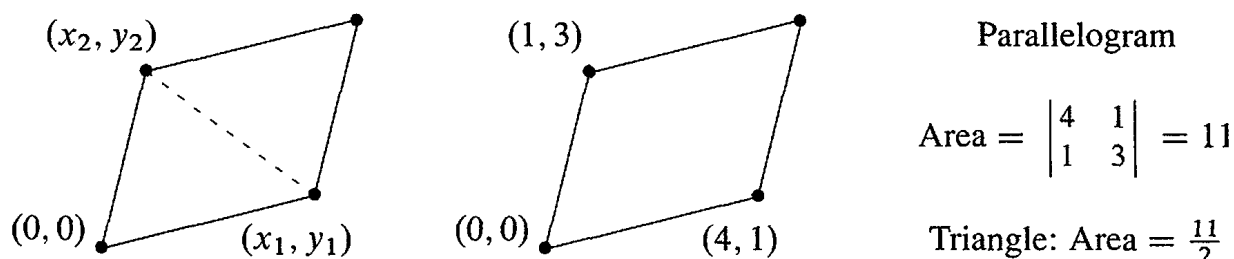


Figure 5.2: A triangle is half of a parallelogram. Area is half of a determinant.

There are many proofs but this one fits with the book. We show that the area has the same properties 1-2-3 as the determinant. Then area = determinant! Remember that those three rules defined the determinant and led to all its other properties.

- 1 When $A = I$, the parallelogram becomes the unit square. Its area is $\det I = 1$.
- 2 When rows are exchanged, the determinant reverses sign. The absolute value (positive area) stays the same—it is the same parallelogram.
- 3 If row 1 is multiplied by t , Figure 5.3a shows that the area is also multiplied by t . Suppose a new row (x'_1, y'_1) is added to (x_1, y_1) (keeping row 2 fixed). Figure 5.3b shows that the solid parallelogram areas add to the dotted parallelogram area (because the two triangles completed by dotted lines are the same).

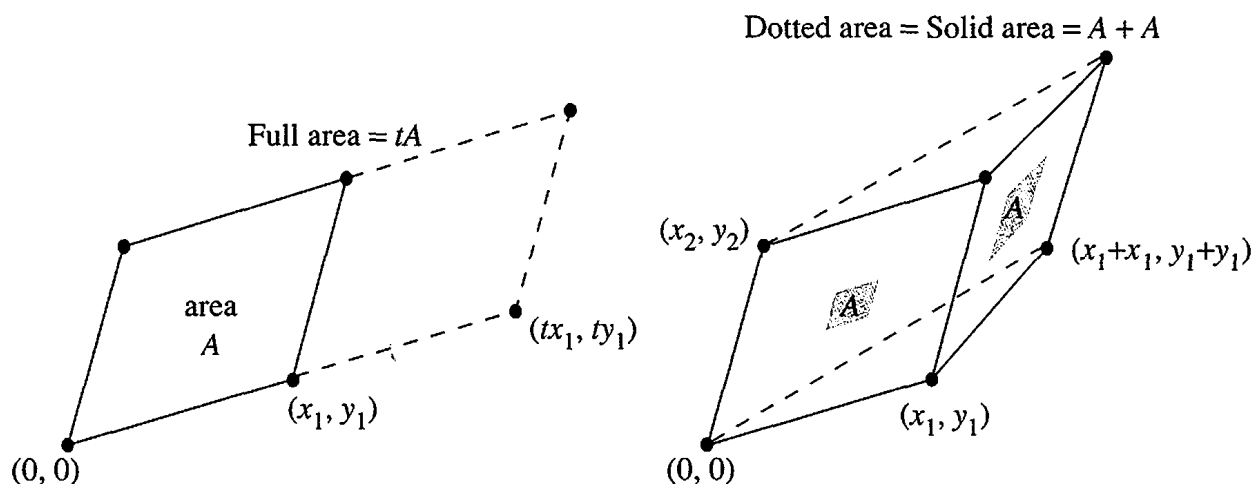


Figure 5.3: Areas obey the rule of linearity (keeping the side (x_2, y_2) constant).

That is an exotic proof, when we could use plane geometry. But the proof has a major attraction—it applies in n dimensions. The n edges going out from the origin are given by the rows of an n by n matrix. The box is completed by more edges, just like the parallelogram.

Figure 5.4 shows a three-dimensional box—whose edges are not at right angles. *The volume equals the absolute value of $\det A$.* Our proof checks again that rules 1–3 for

determinants are also obeyed by volumes. When an edge is stretched by a factor t , the volume is multiplied by t . When edge 1 is added to edge 1', the new box has edge $1 + 1'$. Its volume is the sum of the two original volumes. This is Figure 5.3b lifted into three dimensions or n dimensions. I would draw the boxes but this paper is only two-dimensional.

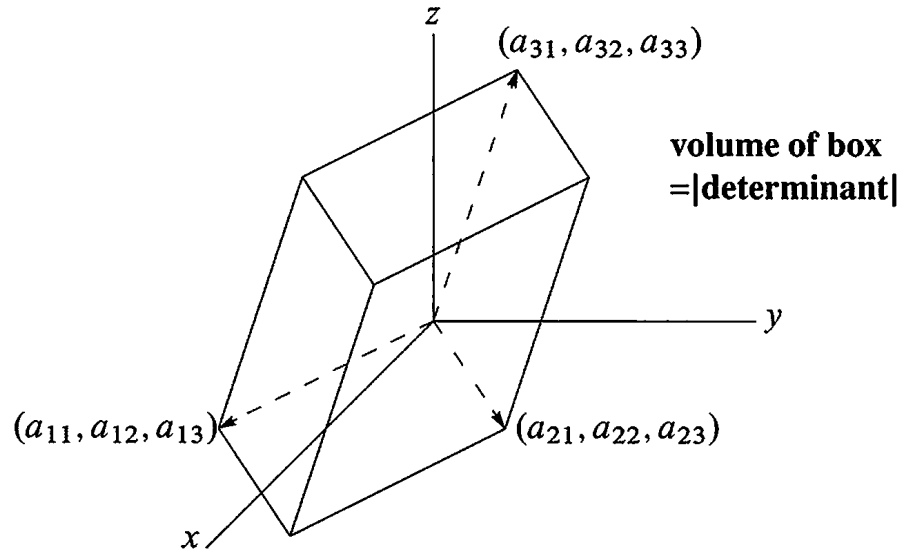


Figure 5.4: Three-dimensional box formed from the three rows of A .

The unit cube has volume $= 1$, which is $\det I$. Row exchanges or edge exchanges leave the same box and the same absolute volume. The determinant changes sign, to indicate whether the edges are a *right-handed triple* ($\det A > 0$) or a *left-handed triple* ($\det A < 0$). The box volume follows the rules for determinants, so volume of the box = absolute value of the determinant.

Example 5 Suppose a rectangular box (90° angles) has side lengths r, s , and t . Its volume is r times s times t . The diagonal matrix with entries r, s , and t produces those three sides. Then $\det A$ also equals $r s t$.

Example 6 In calculus, the box is infinitesimally small! To integrate over a circle, we might change x and y to r and θ . Those are polar coordinates: $x = r \cos \theta$ and $y = r \sin \theta$. The area of a “polar box” is a determinant J times $dr d\theta$:

$$J = \begin{vmatrix} \partial x / \partial r & \partial x / \partial \theta \\ \partial y / \partial r & \partial y / \partial \theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r.$$

This determinant is the r in the small area $dA = r dr d\theta$. The stretching factor J goes into double integrals just as dx/du goes into an ordinary integral $\int dx = \int (dx/du) du$. For triple integrals the Jacobian matrix J with nine derivatives will be 3 by 3.

The Cross Product

The *cross product* is an extra (and optional) application, special for three dimensions. Start with vectors $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$. Unlike the dot product, which is a number, the cross product is a vector—also in three dimensions. It is written $\mathbf{u} \times \mathbf{v}$ and pronounced “ $\mathbf{u}$ cross $\mathbf{v}$.” *The components of this cross product are just 2 by 2 cofactors.* We will explain the properties that make $\mathbf{u} \times \mathbf{v}$ useful in geometry and physics.

This time we bite the bullet, and write down the formula before the properties.

DEFINITION The *cross product* of $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ is a vector

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2v_3 - u_3v_2)\mathbf{i} + (u_3v_1 - u_1v_3)\mathbf{j} + (u_1v_2 - u_2v_1)\mathbf{k}. \quad (10)$$

This vector is perpendicular to $\mathbf{u}$ and $\mathbf{v}$. The cross product $\mathbf{v} \times \mathbf{u}$ is $-(\mathbf{u} \times \mathbf{v})$.

Comment The 3 by 3 determinant is the easiest way to remember $\mathbf{u} \times \mathbf{v}$. It is not especially legal, because the first row contains vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$ and the other rows contain numbers. In the determinant, the vector $\mathbf{i} = (1, 0, 0)$ multiplies u_2v_3 and $-u_3v_2$. The result is $(u_2v_3 - u_3v_2, 0, 0)$, which displays the first component of the cross product.

Notice the cyclic pattern of the subscripts: 2 and 3 give component 1 of $\mathbf{u} \times \mathbf{v}$, then 3 and 1 give component 2, then 1 and 2 give component 3. This completes the definition of $\mathbf{u} \times \mathbf{v}$. Now we list the properties of the cross product:

Property 1 $\mathbf{v} \times \mathbf{u}$ reverses rows 2 and 3 in the determinant so it equals $-(\mathbf{u} \times \mathbf{v})$.

Property 2 The cross product $\mathbf{u} \times \mathbf{v}$ is perpendicular to $\mathbf{u}$ (and also to $\mathbf{v}$). The direct proof is to watch terms cancel. Perpendicularity is a zero dot product:

$$\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = u_1(u_2v_3 - u_3v_2) + u_2(u_3v_1 - u_1v_3) + u_3(u_1v_2 - u_2v_1) = 0. \quad (11)$$

The determinant now has rows $\mathbf{u}, \mathbf{u}$ and $\mathbf{v}$ so it is zero.

Property 3 The cross product of any vector with itself (two equal rows) is $\mathbf{u} \times \mathbf{u} = \mathbf{0}$.

When $\mathbf{u}$ and $\mathbf{v}$ are parallel, the cross product is zero. When $\mathbf{u}$ and $\mathbf{v}$ are perpendicular, the dot product is zero. One involves $\sin \theta$ and the other involves $\cos \theta$:

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| |\sin \theta| \quad \text{and} \quad |\mathbf{u} \cdot \mathbf{v}| = \|\mathbf{u}\| \|\mathbf{v}\| |\cos \theta|. \quad (12)$$

Example 7 Since $\mathbf{u} = (3, 2, 0)$ and $\mathbf{v} = (1, 4, 0)$ are in the xy plane, $\mathbf{u} \times \mathbf{v}$ goes up the z axis:

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 2 & 0 \\ 1 & 4 & 0 \end{vmatrix} = 10\mathbf{k}. \quad \text{The cross product is } \mathbf{u} \times \mathbf{v} = (0, 0, 10).$$

The length of $u \times v$ equals the area of the parallelogram with sides u and v . This will be important: In this example the area is 10.

Example 8 The cross product of $u = (1, 1, 1)$ and $v = (1, 1, 2)$ is $(1, -1, 0)$:

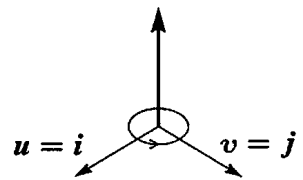
$$\begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{vmatrix} = i \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} - j \begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} + k \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} = i - j.$$

This vector $(1, -1, 0)$ is perpendicular to $(1, 1, 1)$ and $(1, 1, 2)$ as predicted. Area $= \sqrt{2}$.

Example 9 The cross product of $(1, 0, 0)$ and $(0, 1, 0)$ obeys the *right hand rule*. It goes up not down:

$$\begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = k$$

$i \times j = k$



Rule $u \times v$ points along your right thumb when the fingers curl from u to v .

Thus $i \times j = k$. The right hand rule also gives $j \times k = i$ and $k \times i = j$. Note the cyclic order. In the opposite order (anti-cyclic) the thumb is reversed and the cross product goes the other way: $k \times j = -i$ and $i \times k = -j$ and $j \times i = -k$. You see the three plus signs and three minus signs from a 3 by 3 determinant.

The definition of $u \times v$ can be based on vectors instead of their components:

DEFINITION The **cross product** is a vector with length $\|u\| \|v\| |\sin \theta|$. Its direction is perpendicular to u and v . It points “up” or “down” by the right hand rule.

This definition appeals to physicists, who hate to choose axes and coordinates. They see (u_1, u_2, u_3) as the position of a mass and (F_x, F_y, F_z) as a force acting on it. If F is parallel to u , then $u \times F = 0$ —there is no turning. The cross product $u \times F$ is the turning force or *torque*. It points along the turning axis (perpendicular to u and F). Its length $\|u\| \|F\| \sin \theta$ measures the “moment” that produces turning.

Triple Product = Determinant = Volume

Since $u \times v$ is a vector, we can take its dot product with a third vector w . That produces the **triple product** $(u \times v) \cdot w$. It is called a “scalar” triple product, because it is a number. In fact it is a determinant—it gives the volume of the u, v, w box:

$$\text{Triple product} \quad (u \times v) \cdot w = \begin{vmatrix} w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}. \quad (13)$$

We can put w in the top or bottom row. The two determinants are the same because _____ row exchanges go from one to the other. Notice when this determinant is zero:

$$(u \times v) \cdot w = 0 \quad \text{exactly when the vectors } u, v, w \text{ lie in the same plane.}$$

First reason $u \times v$ is perpendicular to that plane so its dot product with w is zero.

Second reason Three vectors in a plane are dependent. The matrix is singular ($\det = 0$).

Third reason Zero volume when the u, v, w box is squashed onto a plane.

It is remarkable that $(u \times v) \cdot w$ equals the volume of the box with sides u, v, w . This 3 by 3 determinant carries tremendous information. Like $ad - bc$ for a 2 by 2 matrix, it separates invertible from singular. Chapter 6 will be looking for singular.

■ REVIEW OF THE KEY IDEAS ■

1. Cramer's Rule solves $Ax = b$ by ratios like $x_1 = |B_1|/|A| = |b \ a_2 \cdots a_n|/|A|$.
2. When C is the cofactor matrix for A , the inverse is $A^{-1} = C^T / \det A$.
3. The volume of a box is $|\det A|$, when the box edges are the rows of A .
4. Area and volume are needed to change variables in double and triple integrals.
5. In $\mathbf{R}^3$, the cross product $u \times v$ is perpendicular to u and v .

■ WORKED EXAMPLES ■

5.3 A If A is singular, the equation $AC^T = (\det A)I$ becomes $AC^T = \text{zero matrix}$. Then each column of C^T is in the nullspace of A . Those columns contain cofactors along rows of A . So the cofactors quickly find the nullspace of a 3 by 3 matrix—my apologies that this comes so late!

Solve $Ax = 0$ by $x = \text{cofactors along a row}$, for these singular matrices of rank 2:

Cofactors give Nullspace	$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 3 & 9 \\ 2 & 2 & 8 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
--------------------------------	---	---

Any nonzero column of C^T will give the desired solution to $Ax = 0$. With rank 2, A has at least one nonzero cofactor. If A has rank 1 we get $x = 0$ and the idea fails.

Solution The first matrix has these cofactors along its top row (note each minus sign):

$$\begin{vmatrix} 3 & 9 \\ 2 & 8 \end{vmatrix} = 6 \qquad - \begin{vmatrix} 2 & 9 \\ 2 & 8 \end{vmatrix} = 2 \qquad \begin{vmatrix} 2 & 3 \\ 2 & 2 \end{vmatrix} = -2$$

Then $\mathbf{x} = (6, 2, -2)$ solves $A\mathbf{x} = \mathbf{0}$. The cofactors along the second row are $(-18, -6, 6)$ which is just $-3\mathbf{x}$. This is also in the one-dimensional nullspace of A .

The second matrix has *zero cofactors* along its first row. The nullvector $\mathbf{x} = (0, 0, 0)$ is not interesting. The cofactors of row 2 give $\mathbf{x} = (1, -1, 0)$ which solves $A\mathbf{x} = \mathbf{0}$.

Every n by n matrix of rank $n - 1$ has at least one nonzero cofactor by Problem 3.3.12. But for rank $n - 2$, all cofactors are zero and we only find $\mathbf{x} = \mathbf{0}$.

5.3 B Use Cramer's Rule with ratios $\det B_j / \det A$ to solve $A\mathbf{x} = \mathbf{b}$. Also find the inverse matrix $A^{-1} = C^T / \det A$. Why is the solution $\mathbf{x}$ for this $\mathbf{b}$ the same as column 3 of A^{-1} ? Which cofactors are involved in computing that column $\mathbf{x}$?

$$A\mathbf{x} = \mathbf{b} \quad \text{is} \quad \begin{bmatrix} 2 & 6 & 2 \\ 1 & 4 & 2 \\ 5 & 9 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Find the volumes of the boxes whose edges are *columns* of A and then rows of A^{-1} .

Solution The determinants of the B_j (with right side $\mathbf{b}$ placed in column j) are

$$|B_1| = \begin{vmatrix} 0 & 6 & 2 \\ 0 & 4 & 2 \\ 1 & 9 & 0 \end{vmatrix} = 4 \qquad |B_2| = \begin{vmatrix} 2 & 0 & 2 \\ 1 & 0 & 2 \\ 5 & 1 & 0 \end{vmatrix} = -2 \qquad |B_3| = \begin{vmatrix} 2 & 6 & 0 \\ 1 & 4 & 0 \\ 5 & 9 & 1 \end{vmatrix} = 2.$$

Those are cofactors C_{31}, C_{32}, C_{33} of row 3. Their dot product with row 3 is $\det A$:

$$\det A = a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} = (5, 9, 0) \cdot (4, -2, 2) = 2.$$

The three ratios $\det B_j / \det A$ give the three components of $\mathbf{x} = (2, -1, 1)$. This $\mathbf{x}$ is the third column of A^{-1} because $\mathbf{b} = (0, 0, 1)$ is the third column of I . The cofactors along the other rows of A , divided by $\det A = 2$, give the other *columns* of A^{-1} :

$$A^{-1} = \frac{C^T}{\det A} = \frac{1}{2} \begin{bmatrix} -18 & 18 & 4 \\ 10 & -10 & -2 \\ -11 & 12 & 2 \end{bmatrix}. \quad \text{Multiply to check} \quad AA^{-1} = I$$

The box from the columns of A has volume $= \det A = 2$ (the same as the box from the rows, since $|A^T| = |A|$). The box from A^{-1} has volume $1/|A| = \frac{1}{2}$.

Problem Set 5.3

Problems 1–5 are about Cramer's Rule for $x = A^{-1}b$.

- 1 Solve these linear equations by Cramer's Rule $x_j = \det B_j / \det A$:

$$\begin{array}{ll} \text{(a)} \quad \begin{array}{l} 2x_1 + 5x_2 = 1 \\ x_1 + 4x_2 = 2 \end{array} & \text{(b)} \quad \begin{array}{l} 2x_1 + x_2 = 1 \\ x_1 + 2x_2 + x_3 = 0 \\ x_2 + 2x_3 = 0. \end{array} \end{array}$$

- 2 Use Cramer's Rule to solve for y (only). Call the 3 by 3 determinant D :

$$\begin{array}{ll} \text{(a)} \quad \begin{array}{l} ax + by = 1 \\ cx + dy = 0 \end{array} & \text{(b)} \quad \begin{array}{l} ax + by + cz = 1 \\ dx + ey + fz = 0 \\ gx + hy + iz = 0. \end{array} \end{array}$$

- 3 Cramer's Rule breaks down when $\det A = 0$. Example (a) has no solution while (b) has infinitely many. What are the ratios $x_j = \det B_j / \det A$ in these two cases?

$$\begin{array}{ll} \text{(a)} \quad \begin{array}{l} 2x_1 + 3x_2 = 1 \\ 4x_1 + 6x_2 = 1 \end{array} \quad (\text{parallel lines}) & \text{(b)} \quad \begin{array}{l} 2x_1 + 3x_2 = 1 \\ 4x_1 + 6x_2 = 2 \end{array} \quad (\text{same line}) \end{array}$$

- 4 *Quick proof of Cramer's rule.* The determinant is a linear function of column 1. It is zero if two columns are equal. When $b = Ax = x_1a_1 + x_2a_2 + x_3a_3$ goes into the first column of A , the determinant of this matrix B_1 is

$$|b \ a_2 \ a_3| = |x_1a_1 + x_2a_2 + x_3a_3 \ a_2 \ a_3| = x_1|a_1 \ a_2 \ a_3| = x_1 \det A.$$

- (a) What formula for x_1 comes from left side = right side?
(b) What steps lead to the middle equation?

- 5 If the right side b is the first column of A , solve the 3 by 3 system $Ax = b$. How does each determinant in Cramer's Rule lead to this solution x ?

Problems 6–15 are about $A^{-1} = C^T / \det A$. Remember to transpose C .

- 6 Find A^{-1} from the cofactor formula $C^T / \det A$. Use symmetry in part (b).

$$\begin{array}{ll} \text{(a)} \quad A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 7 & 1 \end{bmatrix} & \text{(b)} \quad A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}. \end{array}$$

- 7 If all the cofactors are zero, how do you know that A has no inverse? If none of the cofactors are zero, is A sure to be invertible?

- 8 Find the cofactors of A and multiply AC^T to find $\det A$:

$$A = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 6 & -3 & 0 \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{bmatrix} \quad \text{and} \quad AC^T = \underline{\hspace{2cm}}.$$

If you change that 4 to 100, why is $\det A$ unchanged?

- 9 Suppose $\det A = 1$ and you know all the cofactors in C . How can you find A ?
- 10 From the formula $AC^T = (\det A)I$ show that $\det C = (\det A)^{n-1}$.
- 11 If all entries of A are integers, and $\det A = 1$ or -1 , prove that all entries of A^{-1} are integers. Give a 2 by 2 example with no zero entries.
- 12 If all entries of A and A^{-1} are integers, prove that $\det A = 1$ or -1 . Hint: What is $\det A$ times $\det A^{-1}$?
- 13 Complete the calculation of A^{-1} by cofactors that was started in Example 5.
- 14 L is lower triangular and S is symmetric. Assume they are invertible:

$$\begin{array}{l} \text{To invert} \\ \text{triangular } L \\ \text{symmetric } S \end{array} \quad L = \begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{bmatrix} \quad S = \begin{bmatrix} a & b & d \\ b & c & e \\ d & e & f \end{bmatrix}.$$

- (a) Which three cofactors of L are zero? Then L^{-1} is also lower triangular.
- (b) Which three pairs of cofactors of S are equal? Then S^{-1} is also symmetric.
- (c) The cofactor matrix C of an orthogonal Q will be _____. Why?
- 15 For $n = 5$ the matrix C contains _____ cofactors. Each 4 by 4 cofactor contains _____ terms and each term needs _____ multiplications. Compare with $5^3 = 125$ for the Gauss-Jordan computation of A^{-1} in Section 2.4.

Problems 16–26 are about area and volume by determinants.

- 16 (a) Find the area of the parallelogram with edges $v = (3, 2)$ and $w = (1, 4)$.
 (b) Find the area of the triangle with sides v , w , and $v + w$. Draw it.
 (c) Find the area of the triangle with sides v , w , and $w - v$. Draw it.
- 17 A box has edges from $(0, 0, 0)$ to $(3, 1, 1)$ and $(1, 3, 1)$ and $(1, 1, 3)$. Find its volume. Also find the area of each parallelogram face using $\|u \times v\|$.
- 18 (a) The corners of a triangle are $(2, 1)$ and $(3, 4)$ and $(0, 5)$. What is the area?
 (b) Add a corner at $(-1, 0)$ to make a lopsided region (four sides). Find the area.
- 19 The parallelogram with sides $(2, 1)$ and $(2, 3)$ has the same area as the parallelogram with sides $(2, 2)$ and $(1, 3)$. Find those areas from 2 by 2 determinants and say why they must be equal. (I can't see why from a picture. Please write to me if you do.)
- 20 The Hadamard matrix H has orthogonal rows. The box is a hypercube!

$$\text{What is } |H| = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{vmatrix} = \text{volume of a hypercube in } \mathbf{R}^4?$$

- 21 If the columns of a 4 by 4 matrix have lengths L_1, L_2, L_3, L_4 , what is the largest possible value for the determinant (based on volume)? If all entries of the matrix are 1 or -1 , what are those lengths and the maximum determinant?
- 22 Show by a picture how a rectangle with area $x_1 y_2$ minus a rectangle with area $x_2 y_1$ produces the same area as our parallelogram.
- 23 When the edge vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are perpendicular, the volume of the box is $\|\mathbf{a}\|$ times $\|\mathbf{b}\|$ times $\|\mathbf{c}\|$. The matrix $A^T A$ is _____. Find $\det A^T A$ and $\det A$.
- 24 The box with edges $\mathbf{i}$ and $\mathbf{j}$ and $\mathbf{w} = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ has height _____. What is the volume? What is the matrix with this determinant? What is $\mathbf{i} \times \mathbf{j}$ and what is its dot product with $\mathbf{w}$?
- 25 An n -dimensional cube has how many corners? How many edges? How many $(n - 1)$ -dimensional faces? The cube in $\mathbf{R}^n$ whose edges are the rows of $2I$ has volume _____. A hypercube computer has parallel processors at the corners with connections along the edges.
- 26 The triangle with corners $(0, 0), (1, 0), (0, 1)$ has area $\frac{1}{2}$. The pyramid in $\mathbf{R}^3$ with four corners $(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)$ has volume _____. What is the volume of a pyramid in $\mathbf{R}^4$ with five corners at $(0, 0, 0, 0)$ and the rows of I ?

Problems 27–30 are about areas dA and volumes dV in calculus.

- 27 Polar coordinates satisfy $x = r \cos \theta$ and $y = r \sin \theta$. Polar area is $J dr d\theta$:

$$J = \begin{vmatrix} \partial x / \partial r & \partial x / \partial \theta \\ \partial y / \partial r & \partial y / \partial \theta \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}.$$

The two columns are orthogonal. Their lengths are _____. Thus $J =$ _____.

- 28 Spherical coordinates ρ, ϕ, θ satisfy $x = \rho \sin \phi \cos \theta$ and $y = \rho \sin \phi \sin \theta$ and $z = \rho \cos \phi$. Find the 3 by 3 matrix of partial derivatives: $\partial x / \partial \rho, \partial x / \partial \phi, \partial x / \partial \theta$ in row 1. Simplify its determinant to $J = \rho^2 \sin \phi$. Then dV in spherical coordinates is $\rho^2 \sin \phi d\rho d\phi d\theta$, the volume of an infinitesimal “coordinate box”.
- 29 The matrix that connects r, θ to x, y is in Problem 27. Invert that 2 by 2 matrix:

$$J^{-1} = \begin{vmatrix} \partial r / \partial x & \partial r / \partial y \\ \partial \theta / \partial x & \partial \theta / \partial y \end{vmatrix} = \begin{vmatrix} \cos \theta & ? \\ ? & ? \end{vmatrix} = ?$$

It is surprising that $\partial r / \partial x = \partial x / \partial r$ (*Calculus*, Gilbert Strang, p. 501). Multiplying the matrices J and J^{-1} gives the chain rule $\frac{\partial x}{\partial x} = \frac{\partial x}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial x}{\partial \theta} \frac{\partial \theta}{\partial x} = 1$.

- 30 The triangle with corners $(0, 0), (6, 0),$ and $(1, 4)$ has area _____. When you rotate it by $\theta = 60^\circ$ the area is _____. The determinant of the rotation matrix is

$$J = \begin{vmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & ? \\ ? & ? \end{vmatrix} = ?$$

Problems 31–38 are about the triple product $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ in three dimensions.

- 31** A box has base area $\|\mathbf{u} \times \mathbf{v}\|$. Its perpendicular height is $\|\mathbf{w}\| \cos \theta$. Base area times height = volume = $\|\mathbf{u} \times \mathbf{v}\| \|\mathbf{w}\| \cos \theta$ which is $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$. Compute base area, height, and volume for $\mathbf{u} = (2, 4, 0)$, $\mathbf{v} = (-1, 3, 0)$, $\mathbf{w} = (1, 2, 2)$.
- 32** The volume of the same box is given more directly by a 3 by 3 determinant. Evaluate that determinant.
- 33** Expand the 3 by 3 determinant in equation (13) in cofactors of its row u_1, u_2, u_3 . This expansion is the dot product of $\mathbf{u}$ with the vector _____.
- 34** Which of the triple products $(\mathbf{u} \times \mathbf{w}) \cdot \mathbf{v}$ and $(\mathbf{w} \times \mathbf{u}) \cdot \mathbf{v}$ and $(\mathbf{v} \times \mathbf{w}) \cdot \mathbf{u}$ are the same as $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$? Which orders of the rows $\mathbf{u}, \mathbf{v}, \mathbf{w}$ give the correct determinant?
- 35** Let $P = (1, 0, -1)$ and $Q = (1, 1, 1)$ and $R = (2, 2, 1)$. Choose S so that $PQRS$ is a parallelogram and compute its area. Choose T, U, V so that $OPQRSTUV$ is a tilted box and compute its volume.
- 36** Suppose (x, y, z) and $(1, 1, 0)$ and $(1, 2, 1)$ lie on a plane through the origin. What determinant is zero? What equation does this give for the plane?
- 37** Suppose (x, y, z) is a linear combination of $(2, 3, 1)$ and $(1, 2, 3)$. What determinant is zero? What equation does this give for the plane of all combinations?
- 38** (a) Explain from volumes why $\det 2A = 2^n \det A$ for n by n matrices.
(b) For what size matrix is the false statement $\det A + \det A = \det(A + A)$ true?

Challenge Problems

- 39** If you know all 16 cofactors of a 4 by 4 invertible matrix A , how would you find A ?
- 40** Suppose A is a 5 by 5 matrix. Its entries in row 1 multiply determinants (cofactors) in rows 2–5 to give the determinant. Can you guess a “Jacobi formula” for $\det A$ using 2 by 2 determinants from rows 1–2 *times* 3 by 3 determinants from rows 3–5? Test your formula on the $-1, 2, -1$ tridiagonal matrix that has determinant = 6.
- 41** The 2 by 2 matrix $AB = (2 \text{ by } 3)(3 \text{ by } 2)$ has a “Cauchy-Binet formula” for $\det AB$:
- $$\det AB = \text{sum of (2 by 2 determinants in } A) (2 \text{ by 2 determinants in } B)$$
- (a) Guess which 2 by 2 determinants to use from A and B .
(b) Test your formula when the rows of A are 1, 2, 3 and 1, 4, 7 with $B = A^T$.

Chapter 6

Eigenvalues and Eigenvectors

6.1 Introduction to Eigenvalues

Linear equations $Ax = b$ come from steady state problems. Eigenvalues have their greatest importance in *dynamic problems*. The solution of $du/dt = Au$ is changing with time—growing or decaying or oscillating. We can't find it by elimination. This chapter enters a new part of linear algebra, based on $Ax = \lambda x$. All matrices in this chapter are square.

A good model comes from the powers $A, A^2, A^3, \dots$ of a matrix. Suppose you need the hundredth power A^{100} . The starting matrix A becomes unrecognizable after a few steps, and A^{100} is very close to $\begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}$:

$$\begin{array}{ccc} \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} & \begin{bmatrix} .70 & .45 \\ .30 & .55 \end{bmatrix} & \begin{bmatrix} .650 & .525 \\ .350 & .475 \end{bmatrix} & \cdots & \begin{bmatrix} .6000 & .6000 \\ .4000 & .4000 \end{bmatrix} \\ A & A^2 & A^3 & & A^{100} \end{array}$$

A^{100} was found by using the *eigenvalues* of A , not by multiplying 100 matrices. Those eigenvalues (here they are 1 and $1/2$) are a new way to see into the heart of a matrix.

To explain eigenvalues, we first explain eigenvectors. Almost all vectors change direction, when they are multiplied by A . ***Certain exceptional vectors x are in the same direction as Ax . Those are the “eigenvectors”.*** Multiply an eigenvector by A , and the vector Ax is a number λ times the original x .

The basic equation is $Ax = \lambda x$. The number λ is an eigenvalue of A .

The eigenvalue λ tells whether the special vector x is stretched or shrunk or reversed or left unchanged—when it is multiplied by A . We may find $\lambda = 2$ or $\frac{1}{2}$ or -1 or 1 . The eigenvalue λ could be zero! Then $Ax = 0x$ means that this eigenvector x is in the nullspace.

If A is the identity matrix, every vector has $Ax = x$. All vectors are eigenvectors of I . All eigenvalues “lambda” are $\lambda = 1$. This is unusual to say the least. Most 2 by 2 matrices have *two* eigenvector directions and *two* eigenvalues. We will show that $\det(A - \lambda I) = 0$.

This section will explain how to compute the x 's and λ 's. It can come early in the course because we only need the determinant of a 2 by 2 matrix. Let me use $\det(A - \lambda I) = 0$ to find the eigenvalues for this first example, and then derive it properly in equation (3).

Example 1 The matrix A has two eigenvalues $\lambda = 1$ and $\lambda = 1/2$. Look at $\det(A - \lambda I)$:

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \quad \det \begin{bmatrix} .8 - \lambda & .3 \\ .2 & .7 - \lambda \end{bmatrix} = \lambda^2 - \frac{3}{2}\lambda + \frac{1}{2} = (\lambda - 1) \left(\lambda - \frac{1}{2} \right).$$

I factored the quadratic into $\lambda - 1$ times $\lambda - \frac{1}{2}$, to see the two eigenvalues $\lambda = 1$ and $\lambda = \frac{1}{2}$. For those numbers, the matrix $A - \lambda I$ becomes *singular* (zero determinant). The eigenvectors x_1 and x_2 are in the nullspaces of $A - I$ and $A - \frac{1}{2}I$.

$(A - I)x_1 = 0$ is $Ax_1 = x_1$ and the first eigenvector is $(.6, .4)$.

$(A - \frac{1}{2}I)x_2 = 0$ is $Ax_2 = \frac{1}{2}x_2$ and the second eigenvector is $(1, -1)$:

$$x_1 = \begin{bmatrix} .6 \\ .4 \end{bmatrix} \quad \text{and} \quad Ax_1 = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} .6 \\ .4 \end{bmatrix} = x_1 \quad (Ax = x \text{ means that } \lambda_1 = 1)$$

$$x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{and} \quad Ax_2 = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} .5 \\ -.5 \end{bmatrix} \quad (\text{this is } \frac{1}{2}x_2 \text{ so } \lambda_2 = \frac{1}{2}).$$

If x_1 is multiplied again by A , we still get x_1 . Every power of A will give $A^n x_1 = x_1$. Multiplying x_2 by A gave $\frac{1}{2}x_2$, and if we multiply again we get $(\frac{1}{2})^2$ times x_2 .

When A is squared, the eigenvectors stay the same. The eigenvalues are squared.

This pattern keeps going, because the eigenvectors stay in their own directions (Figure 6.1) and never get mixed. The eigenvectors of A^{100} are the same x_1 and x_2 . The eigenvalues of A^{100} are $1^{100} = 1$ and $(\frac{1}{2})^{100} = \text{very small number}$.

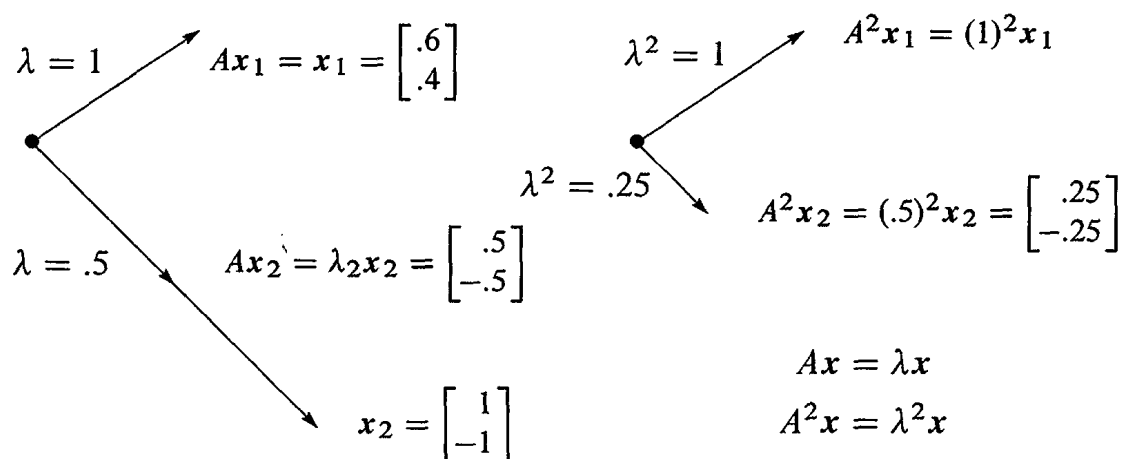


Figure 6.1: The eigenvectors keep their directions. A^2 has eigenvalues 1^2 and $(.5)^2$.

Other vectors do change direction. But all other vectors are combinations of the two eigenvectors. The first column of A is the combination $x_1 + (.2)x_2$:

$$\text{Separate into eigenvectors} \quad \begin{bmatrix} .8 \\ .2 \end{bmatrix} = x_1 + (.2)x_2 = \begin{bmatrix} .6 \\ .4 \end{bmatrix} + \begin{bmatrix} .2 \\ -.2 \end{bmatrix}. \quad (1)$$

Multiplying by A gives $(.7, .3)$, the first column of A^2 . Do it separately for x_1 and $(.2)x_2$. Of course $Ax_1 = x_1$. And A multiplies x_2 by its eigenvalue $\frac{1}{2}$:

Multiply each x_i by λ_i $A \begin{bmatrix} .8 \\ .2 \end{bmatrix} = \begin{bmatrix} .7 \\ .3 \end{bmatrix}$ is $x_1 + \frac{1}{2}(.2)x_2 = \begin{bmatrix} .6 \\ .4 \end{bmatrix} + \begin{bmatrix} .1 \\ -.1 \end{bmatrix}$.

Each eigenvector is multiplied by its eigenvalue, when we multiply by A . We didn't need these eigenvectors to find A^2 . But it is the good way to do 99 multiplications. At every step x_1 is unchanged and x_2 is multiplied by $(\frac{1}{2})$, so we have $(\frac{1}{2})^{99}$:

$$A^{99} \begin{bmatrix} .8 \\ .2 \end{bmatrix} \text{ is really } x_1 + (.2)(\frac{1}{2})^{99}x_2 = \begin{bmatrix} .6 \\ .4 \end{bmatrix} + \begin{bmatrix} \text{very} \\ \text{small} \\ \text{vector} \end{bmatrix}.$$

This is the first column of A^{100} . The number we originally wrote as .6000 was not exact. We left out $(.2)(\frac{1}{2})^{99}$ which wouldn't show up for 30 decimal places.

The eigenvector x_1 is a “steady state” that doesn't change (because $\lambda_1 = 1$). The eigenvector x_2 is a “decaying mode” that virtually disappears (because $\lambda_2 = .5$). The higher the power of A , the closer its columns approach the steady state.

We mention that this particular A is a **Markov matrix**. Its entries are positive and every column adds to 1. Those facts guarantee that the largest eigenvalue is $\lambda = 1$ (as we found). Its eigenvector $x_1 = (.6, .4)$ is the *steady state*—which all columns of A^k will approach. Section 8.3 shows how Markov matrices appear in applications like Google.

For projections we can spot the steady state ($\lambda = 1$) and the nullspace ($\lambda = 0$).

Example 2 The projection matrix $P = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$ has eigenvalues $\lambda = 1$ and $\lambda = 0$.

Its eigenvectors are $x_1 = (1, 1)$ and $x_2 = (1, -1)$. For those vectors, $Px_1 = x_1$ (steady state) and $Px_2 = 0$ (nullspace). This example illustrates Markov matrices and singular matrices and (most important) symmetric matrices. All have special λ 's and x 's:

1. Each column of $P = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$ adds to 1, so $\lambda = 1$ is an eigenvalue.
2. P is **singular**, so $\lambda = 0$ is an eigenvalue.
3. P is **symmetric**, so its eigenvectors $(1, 1)$ and $(1, -1)$ are perpendicular.

The only eigenvalues of a projection matrix are 0 and 1. The eigenvectors for $\lambda = 0$ (which means $Px = 0x$) fill up the nullspace. The eigenvectors for $\lambda = 1$ (which means $Px = x$) fill up the column space. The nullspace is projected to zero. The column space projects onto itself. The projection keeps the column space and destroys the nullspace:

Project each part $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ projects onto $Pv = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

Special properties of a matrix lead to special eigenvalues and eigenvectors. That is a major theme of this chapter (it is captured in a table at the very end).

Projections have $\lambda = 0$ and 1. Permutations have all $|\lambda| = 1$. The next matrix R (a reflection and at the same time a permutation) is also special.

Example 3 The reflection matrix $R = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has eigenvalues 1 and -1 .

The eigenvector $(1, 1)$ is unchanged by R . The second eigenvector is $(1, -1)$ —its signs are reversed by R . A matrix with no negative entries can still have a negative eigenvalue! The eigenvectors for R are the same as for P , because $\text{reflection} = 2(\text{projection}) - I$:

$$R = 2P - I \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 2 \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (2)$$

Here is the point. If $Px = \lambda x$ then $2Px = 2\lambda x$. The eigenvalues are doubled when the matrix is doubled. Now subtract $Ix = x$. The result is $(2P - I)x = (2\lambda - 1)x$. **When a matrix is shifted by I , each λ is shifted by 1.** No change in eigenvectors.

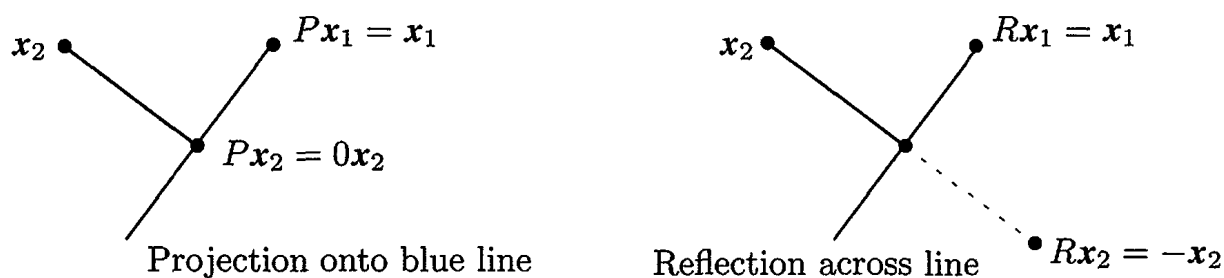


Figure 6.2: Projections P have eigenvalues 1 and 0. Reflections R have $\lambda = 1$ and -1 . A typical x changes direction, but not the eigenvectors x_1 and x_2 .

Key idea: The eigenvalues of R and P are related exactly as the matrices are related:

The eigenvalues of $R = 2P - I$ are $2(1) - 1 = 1$ and $2(0) - 1 = -1$.

The eigenvalues of R^2 are λ^2 . In this case $R^2 = I$. Check $(1)^2 = 1$ and $(-1)^2 = 1$.

The Equation for the Eigenvalues

For projections and reflections we found λ 's and x 's by geometry: $Px = x$, $Px = 0$, $Rx = -x$. Now we use determinants and linear algebra. *This is the key calculation in the chapter*—almost every application starts by solving $Ax = \lambda x$.

First move λx to the left side. Write the equation $Ax = \lambda x$ as $(A - \lambda I)x = 0$. The matrix $A - \lambda I$ times the eigenvector x is the zero vector. **The eigenvectors make up the nullspace of $A - \lambda I$.** When we know an eigenvalue λ , we find an eigenvector by solving $(A - \lambda I)x = 0$.

Eigenvalues first. If $(A - \lambda I)x = 0$ has a nonzero solution, $A - \lambda I$ is not invertible. **The determinant of $A - \lambda I$ must be zero.** This is how to recognize an eigenvalue λ :

Eigenvalues The number λ is an eigenvalue of A if and only if $A - \lambda I$ is singular:

Equation for the eigenvalues $\det(A - \lambda I) = 0.$ (3)

This “characteristic polynomial” $\det(A - \lambda I)$ involves only λ , not x . When A is n by n , equation (3) has degree n . Then A has n eigenvalues (repeats possible!) Each λ leads to x :

For each eigenvalue λ solve $(A - \lambda I)x = 0$ or $Ax = \lambda x$ to find an eigenvector x .

Example 4 $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is already singular (zero determinant). Find its λ 's and x 's.

When A is singular, $\lambda = 0$ is one of the eigenvalues. The equation $Ax = 0x$ has solutions. They are the eigenvectors for $\lambda = 0$. But $\det(A - \lambda I) = 0$ is the way to find *all* λ 's and x 's. Always subtract λI from A :

Subtract λ from the diagonal to find $A - \lambda I = \begin{bmatrix} 1-\lambda & 2 \\ 2 & 4-\lambda \end{bmatrix}.$ (4)

Take the determinant “ $ad - bc$ ” of this 2 by 2 matrix. From $1 - \lambda$ times $4 - \lambda$, the “ ad ” part is $\lambda^2 - 5\lambda + 4$. The “ bc ” part, not containing λ , is 2 times 2.

$$\det \begin{bmatrix} 1-\lambda & 2 \\ 2 & 4-\lambda \end{bmatrix} = (1-\lambda)(4-\lambda) - (2)(2) = \lambda^2 - 5\lambda. \quad (5)$$

Set this determinant $\lambda^2 - 5\lambda$ to zero. One solution is $\lambda = 0$ (as expected, since A is singular). Factoring into λ times $\lambda - 5$, the other root is $\lambda = 5$:

$$\det(A - \lambda I) = \lambda^2 - 5\lambda = 0 \quad \text{yields the eigenvalues} \quad \lambda_1 = 0 \quad \text{and} \quad \lambda_2 = 5.$$

Now find the eigenvectors. Solve $(A - \lambda I)x = 0$ separately for $\lambda_1 = 0$ and $\lambda_2 = 5$:

$$\begin{aligned} (A - 0I)x &= \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{yields an eigenvector} \quad \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad \text{for } \lambda_1 = 0 \\ (A - 5I)x &= \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{yields an eigenvector} \quad \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{for } \lambda_2 = 5. \end{aligned}$$

The matrices $A - 0I$ and $A - 5I$ are singular (because 0 and 5 are eigenvalues). The eigenvectors $(2, -1)$ and $(1, 2)$ are in the nullspaces: $(A - \lambda I)x = 0$ is $Ax = \lambda x$.

We need to emphasize: *There is nothing exceptional about $\lambda = 0$.* Like every other number, zero might be an eigenvalue and it might not. If A is singular, it is. The eigenvectors fill the nullspace: $Ax = 0x = 0$. If A is invertible, zero is not an eigenvalue. We shift A by a multiple of I to make it singular.

In the example, the shifted matrix $A - 5I$ is singular and 5 is the other eigenvalue.

Summary To solve the eigenvalue problem for an n by n matrix, follow these steps:

1. **Compute the determinant of $A - \lambda I$.** With λ subtracted along the diagonal, this determinant starts with λ^n or $-\lambda^n$. It is a polynomial in λ of degree n .
2. **Find the roots of this polynomial,** by solving $\det(A - \lambda I) = 0$. The n roots are the n eigenvalues of A . They make $A - \lambda I$ singular.
3. For each eigenvalue λ , **solve $(A - \lambda I)x = 0$ to find an eigenvector x .**

A note on the eigenvectors of 2 by 2 matrices. When $A - \lambda I$ is singular, both rows are multiples of a vector (a, b) . The eigenvector is any multiple of $(b, -a)$. The example had $\lambda = 0$ and $\lambda = 5$:

$\lambda = 0$: rows of $A - 0I$ in the direction $(1, 2)$; eigenvector in the direction $(2, -1)$

$\lambda = 5$: rows of $A - 5I$ in the direction $(-4, 2)$; eigenvector in the direction $(2, 4)$.

Previously we wrote that last eigenvector as $(1, 2)$. Both $(1, 2)$ and $(2, 4)$ are correct. There is a whole *line of eigenvectors*—any nonzero multiple of x is as good as x . MATLAB's `eig(A)` divides by the length, to make the eigenvector into a unit vector.

We end with a warning. Some 2 by 2 matrices have only *one* line of eigenvectors. This can only happen when two eigenvalues are equal. (On the other hand $A = I$ has equal eigenvalues and plenty of eigenvectors.) Similarly some n by n matrices don't have n independent eigenvectors. Without n eigenvectors, we don't have a basis. We can't write every v as a combination of eigenvectors. In the language of the next section, we can't diagonalize a matrix without n independent eigenvectors.

Good News, Bad News

Bad news first: If you add a row of A to another row, or exchange rows, the eigenvalues usually change. *Elimination does not preserve the λ 's*. The triangular U has *its* eigenvalues sitting along the diagonal—they are the pivots. But they are not the eigenvalues of A ! Eigenvalues are changed when row 1 is added to row 2:

$$U = \begin{bmatrix} 1 & 3 \\ 0 & 0 \end{bmatrix} \quad \text{has } \lambda = 0 \text{ and } \lambda = 1; \quad A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \quad \text{has } \lambda = 0 \text{ and } \lambda = 7.$$

Good news second: The *product λ_1 times λ_2 and the sum $\lambda_1 + \lambda_2$ can be found quickly from the matrix*. For this A , the product is 0 times 7. That agrees with the determinant (which is 0). The sum of eigenvalues is 0 + 7. That agrees with the sum down the main diagonal (the **trace** is 1 + 6). These quick checks always work:

*The product of the n eigenvalues equals the determinant.
The sum of the n eigenvalues equals the sum of the n diagonal entries.*

The sum of the entries on the main diagonal is called the *trace* of A :

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = \text{trace} = a_{11} + a_{22} + \cdots + a_{nn}. \quad (6)$$

Those checks are very useful. They are proved in Problems 16–17 and again in the next section. They don't remove the pain of computing λ 's. But when the computation is wrong, they generally tell us so. To compute the correct λ 's, go back to $\det(A - \lambda I) = 0$.

The determinant test makes the *product* of the λ 's equal to the *product* of the pivots (assuming no row exchanges). But the sum of the λ 's is not the sum of the pivots—as the example showed. The individual λ 's have almost nothing to do with the pivots. In this new part of linear algebra, the key equation is really *nonlinear*: λ multiplies x .

Why do the eigenvalues of a triangular matrix lie on its diagonal?

Imaginary Eigenvalues

One more bit of news (not too terrible). The eigenvalues might not be real numbers.

Example 5 *The 90° rotation $Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ has no real eigenvectors. Its eigenvalues are $\lambda = i$ and $\lambda = -i$. Sum of λ 's = trace = 0. Product = determinant = 1.*

After a rotation, *no vector* Qx stays in the same direction as x (except $x = 0$ which is useless). There cannot be an eigenvector, unless we go to *imaginary numbers*. Which we do.

To see how i can help, look at Q^2 which is $-I$. If Q is rotation through 90°, then Q^2 is rotation through 180°. Its eigenvalues are -1 and -1 . (Certainly $-Ix = -1x$.) Squaring Q will square each λ , so we must have $\lambda^2 = -1$. *The eigenvalues of the 90° rotation matrix Q are $+i$ and $-i$, because $i^2 = -1$.*

Those λ 's come as usual from $\det(Q - \lambda I) = 0$. This equation gives $\lambda^2 + 1 = 0$. Its roots are i and $-i$. We meet the imaginary number i also in the eigenvectors:

$$\begin{array}{l} \text{Complex} \\ \text{eigenvectors} \end{array} \quad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = -i \begin{bmatrix} 1 \\ i \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} i \\ 1 \end{bmatrix} = i \begin{bmatrix} i \\ 1 \end{bmatrix}.$$

Somehow these complex vectors $x_1 = (1, i)$ and $x_2 = (i, 1)$ keep their direction as they are rotated. Don't ask me how. This example makes the all-important point that real matrices can easily have complex eigenvalues and eigenvectors. The particular eigenvalues i and $-i$ also illustrate two special properties of Q :

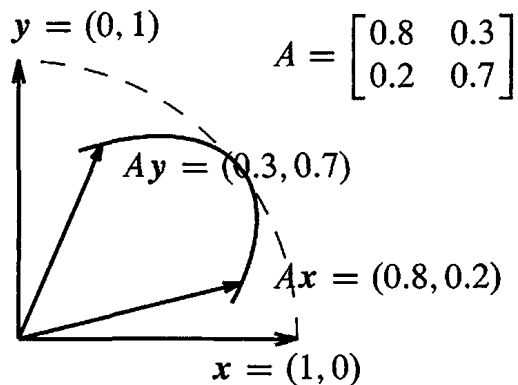
1. Q is an orthogonal matrix so the absolute value of each λ is $|\lambda| = 1$.
2. Q is a skew-symmetric matrix so each λ is pure imaginary.

A symmetric matrix ($A^T = A$) can be compared to a real number. A skew-symmetric matrix ($A^T = -A$) can be compared to an imaginary number. An orthogonal matrix ($A^T A = I$) can be compared to a complex number with $|\lambda| = 1$. For the eigenvalues those are more than analogies—they are theorems to be proved in Section 6.4.

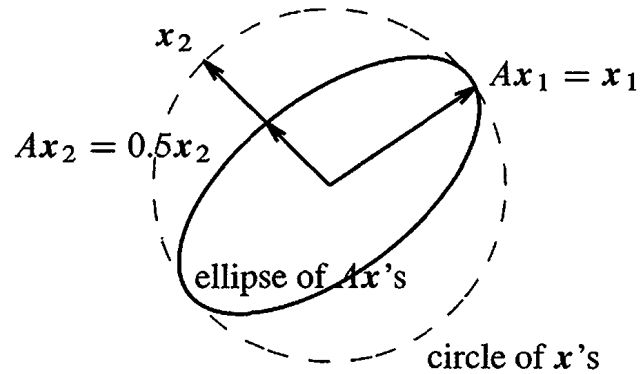
The eigenvectors for all these special matrices are perpendicular. Somehow $(i, 1)$ and $(1, i)$ are perpendicular (Chapter 10 explains the dot product of complex vectors).

Eigshow in MATLAB

There is a MATLAB demo (just type **eigshow**), displaying the eigenvalue problem for a 2 by 2 matrix. It starts with the unit vector $x = (1, 0)$. *The mouse makes this vector move around the unit circle.* At the same time the screen shows Ax , in color and also moving. Possibly Ax is ahead of x . Possibly Ax is behind x . *Sometimes Ax is parallel to x .* At that parallel moment, $Ax = \lambda x$ (at x_1 and x_2 in the second figure).



These are not eigenvectors



Ax lines up with x at eigenvectors

The eigenvalue λ is the length of Ax , when the unit eigenvector x lines up. The built-in choices for A illustrate three possibilities: 0, 1, or 2 directions where Ax crosses x .

0. There are *no real eigenvectors*. Ax stays behind or ahead of x . This means the eigenvalues and eigenvectors are complex, as they are for the rotation Q .
1. There is only *one* line of eigenvectors (unusual). The moving directions Ax and x touch but don't cross over. This happens for the last 2 by 2 matrix below.
2. There are eigenvectors in *two* independent directions. This is typical! Ax crosses x at the first eigenvector x_1 , and it crosses back at the second eigenvector x_2 . Then Ax and x cross again at $-x_1$ and $-x_2$.

You can mentally follow x and Ax for these five matrices. Under the matrices I will count their real eigenvectors. Can you see where Ax lines up with x ?

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

2 2 0 1 1

When A is singular (rank one), its column space is a line. The vector Ax goes up and down that line while x circles around. One eigenvector x is along the line. Another eigenvector appears when $Ax_2 = \mathbf{0}$. Zero is an eigenvalue of a singular matrix.

■ REVIEW OF THE KEY IDEAS ■

1. $Ax = \lambda x$ says that eigenvectors x keep the same direction when multiplied by A .
2. $Ax = \lambda x$ also says that $\det(A - \lambda I) = 0$. This determines n eigenvalues.
3. The eigenvalues of A^2 and A^{-1} are λ^2 and λ^{-1} , with the same eigenvectors.
4. The sum of the λ 's equals the sum down the main diagonal of A (*the trace*). The product of the λ 's equals the determinant.
5. Projections P , reflections R , 90° rotations Q have special eigenvalues $1, 0, -1, i, -i$. Singular matrices have $\lambda = 0$. Triangular matrices have λ 's on their diagonal.

■ WORKED EXAMPLES ■

6.1 A Find the eigenvalues and eigenvectors of A and A^2 and A^{-1} and $A + 4I$:

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \quad \text{and} \quad A^2 = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}.$$

Check the trace $\lambda_1 + \lambda_2$ and the determinant $\lambda_1 \lambda_2$ for A and also A^2 .

Solution The eigenvalues of A come from $\det(A - \lambda I) = 0$:

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} = \lambda^2 - 4\lambda + 3 = 0.$$

This factors into $(\lambda - 1)(\lambda - 3) = 0$ so the eigenvalues of A are $\lambda_1 = 1$ and $\lambda_2 = 3$. For the trace, the sum $2 + 2$ agrees with $1 + 3$. The determinant 3 agrees with the product $\lambda_1 \lambda_2 = 3$. The eigenvectors come separately by solving $(A - \lambda I)x = \mathbf{0}$ which is $Ax = \lambda x$:

$$\lambda = 1: (A - I)x = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ gives the eigenvector } x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda = 3: (A - 3I)x = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ gives the eigenvector } x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

A^2 and A^{-1} and $A + 4I$ keep the *same eigenvectors* as A . Their eigenvalues are λ^2 and λ^{-1} and $\lambda + 4$:

$$A^2 \text{ has eigenvalues } 1^2 = 1 \text{ and } 3^2 = 9 \quad A^{-1} \text{ has } \frac{1}{1} \text{ and } \frac{1}{3} \quad A + 4I \text{ has } \begin{matrix} 1 + 4 = 5 \\ 3 + 4 = 7 \end{matrix}$$

The trace of A^2 is $5 + 5$ which agrees with $1 + 9$. The determinant is $25 - 16 = 9$.

Notes for later sections: A has *orthogonal eigenvectors* (Section 6.4 on symmetric matrices). A can be *diagonalized* since $\lambda_1 \neq \lambda_2$ (Section 6.2). A is *similar* to any 2 by 2 matrix with eigenvalues 1 and 3 (Section 6.6). A is a *positive definite matrix* (Section 6.5) since $A = A^T$ and the λ 's are positive.

6.1 B Find the eigenvalues and eigenvectors of this 3 by 3 matrix A :

Symmetric matrix

Singular matrix

Trace $1 + 2 + 1 = 4$

$$A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

Solution Since all rows of A add to zero, the vector $x = (1, 1, 1)$ gives $Ax = 0$. This is an eigenvector for the eigenvalue $\lambda = 0$. To find λ_2 and λ_3 I will compute the 3 by 3 determinant:

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & -1 & 0 \\ -1 & 2 - \lambda & -1 \\ 0 & -1 & 1 - \lambda \end{vmatrix} = \begin{aligned} & (1 - \lambda)(2 - \lambda)(1 - \lambda) - 2(1 - \lambda) \\ & = (1 - \lambda)[(2 - \lambda)(1 - \lambda) - 2] \\ & = (1 - \lambda)(-\lambda)(3 - \lambda). \end{aligned}$$

That factor $-\lambda$ confirms that $\lambda = 0$ is a root, and an eigenvalue of A . The other factors $(1 - \lambda)$ and $(3 - \lambda)$ give the other eigenvalues 1 and 3, adding to 4 (the trace). Each eigenvalue 0, 1, 3 corresponds to an eigenvector:

$$x_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad Ax_1 = 0x_1 \quad x_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad Ax_2 = 1x_2 \quad x_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad Ax_3 = 3x_3.$$

I notice again that eigenvectors are perpendicular when A is symmetric.

The 3 by 3 matrix produced a third-degree (cubic) polynomial for $\det(A - \lambda I) = -\lambda^3 + 4\lambda^2 - 3\lambda$. We were lucky to find simple roots $\lambda = 0, 1, 3$. Normally we would use a command like `eig(A)`, and the computation will never even use determinants (Section 9.3 shows a better way for large matrices).

The full command `[S, D] = eig(A)` will produce unit eigenvectors in the columns of the **eigenvector matrix** S . The first one happens to have three minus signs, reversed from $(1, 1, 1)$ and divided by $\sqrt{3}$. The eigenvalues of A will be on the diagonal of the **eigenvalue matrix** (typed as D but soon called Λ).

Problem Set 6.1

- 1 The example at the start of the chapter has powers of this matrix A :

$$A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} \quad \text{and} \quad A^2 = \begin{bmatrix} .70 & .45 \\ .30 & .55 \end{bmatrix} \quad \text{and} \quad A^\infty = \begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}.$$

Find the eigenvalues of these matrices. All powers have the same eigenvectors.

- (a) Show from A how a row exchange can produce different eigenvalues.
- (b) Why is a zero eigenvalue *not* changed by the steps of elimination?

- 2 Find the eigenvalues and the eigenvectors of these two matrices:

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad A + I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}.$$

$A + I$ has the _____ eigenvectors as A . Its eigenvalues are _____ by 1.

- 3 Compute the eigenvalues and eigenvectors of A and A^{-1} . Check the trace !

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} -1/2 & 1 \\ 1/2 & 0 \end{bmatrix}.$$

A^{-1} has the _____ eigenvectors as A . When A has eigenvalues λ_1 and λ_2 , its inverse has eigenvalues _____.

- 4 Compute the eigenvalues and eigenvectors of A and A^2 :

$$A = \begin{bmatrix} -1 & 3 \\ 2 & 0 \end{bmatrix} \quad \text{and} \quad A^2 = \begin{bmatrix} 7 & -3 \\ -2 & 6 \end{bmatrix}.$$

A^2 has the same _____ as A . When A has eigenvalues λ_1 and λ_2 , A^2 has eigenvalues _____. In this example, why is $\lambda_1^2 + \lambda_2^2 = 13$?

- 5 Find the eigenvalues of A and B (easy for triangular matrices) and $A + B$:

$$A = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad A + B = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}.$$

Eigenvalues of $A + B$ (are equal to)(are not equal to) eigenvalues of A plus eigenvalues of B .

- 6 Find the eigenvalues of A and B and AB and BA :

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad AB = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}.$$

- (a) Are the eigenvalues of AB equal to eigenvalues of A times eigenvalues of B ?
- (b) Are the eigenvalues of AB equal to the eigenvalues of BA ?

- 7 Elimination produces $A = LU$. The eigenvalues of U are on its diagonal; they are the _____. The eigenvalues of L are on its diagonal; they are all _____. The eigenvalues of A are not the same as _____.
- 8 (a) If you know that x is an eigenvector, the way to find λ is to _____.
 (b) If you know that λ is an eigenvalue, the way to find x is to _____.
- 9 What do you do to the equation $Ax = \lambda x$, in order to prove (a), (b), and (c)?
 (a) λ^2 is an eigenvalue of A^2 , as in Problem 4.
 (b) λ^{-1} is an eigenvalue of A^{-1} , as in Problem 3.
 (c) $\lambda + 1$ is an eigenvalue of $A + I$, as in Problem 2.
- 10 Find the eigenvalues and eigenvectors for both of these Markov matrices A and A^∞ . Explain from those answers why A^{100} is close to A^∞ :

$$A = \begin{bmatrix} .6 & .2 \\ .4 & .8 \end{bmatrix} \quad \text{and} \quad A^\infty = \begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 2/3 \end{bmatrix}.$$

- 11 Here is a strange fact about 2 by 2 matrices with eigenvalues $\lambda_1 \neq \lambda_2$: The columns of $A - \lambda_1 I$ are multiples of the eigenvector x_2 . Any idea why this should be?
- 12 Find three eigenvectors for this matrix P (projection matrices have $\lambda = 1$ and 0):

Projection matrix $P = \begin{bmatrix} .2 & .4 & 0 \\ .4 & .8 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

If two eigenvectors share the same λ , so do all their linear combinations. Find an eigenvector of P with no zero components.

- 13 From the unit vector $u = (\frac{1}{6}, \frac{1}{6}, \frac{3}{6}, \frac{5}{6})$ construct the rank one projection matrix $P = uu^T$. This matrix has $P^2 = P$ because $u^T u = 1$.
 (a) $Pu = u$ comes from $(uu^T)u = u(\text{_____})$. Then u is an eigenvector with $\lambda = 1$.
 (b) If v is perpendicular to u show that $Pv = 0$. Then $\lambda = 0$.
 (c) Find three independent eigenvectors of P all with eigenvalue $\lambda = 0$.
- 14 Solve $\det(Q - \lambda I) = 0$ by the quadratic formula to reach $\lambda = \cos \theta \pm i \sin \theta$:

$$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \quad \text{rotates the } xy \text{ plane by the angle } \theta. \text{ No real } \lambda\text{'s.}$$

Find the eigenvectors of Q by solving $(Q - \lambda I)x = 0$. Use $i^2 = -1$.

- 15** Every permutation matrix leaves $\mathbf{x} = (1, 1, \dots, 1)$ unchanged. Then $\lambda = 1$. Find two more λ 's (possibly complex) for these permutations, from $\det(P - \lambda I) = 0$:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

- 16** **The determinant of A equals the product $\lambda_1 \lambda_2 \cdots \lambda_n$.** Start with the polynomial $\det(A - \lambda I)$ separated into its n factors (always possible). Then set $\lambda = 0$:

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda) \quad \text{so} \quad \det A = \underline{\hspace{2cm}}.$$

Check this rule in Example 1 where the Markov matrix has $\lambda = 1$ and $\frac{1}{2}$.

- 17** The sum of the diagonal entries (the *trace*) equals the sum of the eigenvalues:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{has} \quad \det(A - \lambda I) = \lambda^2 - (a + d)\lambda + ad - bc = 0.$$

The quadratic formula gives the eigenvalues $\lambda = (a + d + \sqrt{\hspace{1cm}})/2$ and $\lambda = \underline{\hspace{1cm}}$. Their sum is $\underline{\hspace{1cm}}$. If A has $\lambda_1 = 3$ and $\lambda_2 = 4$ then $\det(A - \lambda I) = \underline{\hspace{1cm}}$.

- 18** If A has $\lambda_1 = 4$ and $\lambda_2 = 5$ then $\det(A - \lambda I) = (\lambda - 4)(\lambda - 5) = \lambda^2 - 9\lambda + 20$. Find three matrices that have trace $a + d = 9$ and determinant 20 and $\lambda = 4, 5$.

- 19** A 3 by 3 matrix B is known to have eigenvalues 0, 1, 2. This information is enough to find three of these (give the answers where possible):

- (a) the rank of B
- (b) the determinant of $B^T B$
- (c) the eigenvalues of $B^T B$
- (d) the eigenvalues of $(B^2 + I)^{-1}$.

- 20** Choose the last rows of A and C to give eigenvalues 4, 7 and 1, 2, 3:

Companion matrices
$$A = \begin{bmatrix} 0 & 1 \\ * & * \end{bmatrix} \quad C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ * & * & * \end{bmatrix}.$$

- 21** **The eigenvalues of A equal the eigenvalues of A^T .** This is because $\det(A - \lambda I)$ equals $\det(A^T - \lambda I)$. That is true because $\underline{\hspace{1cm}}$. Show by an example that the eigenvectors of A and A^T are *not* the same.

- 22** Construct any 3 by 3 Markov matrix M : positive entries down each column add to 1. Show that $M^T(1, 1, 1) = (1, 1, 1)$. By Problem 21, $\lambda = 1$ is also an eigenvalue of M . Challenge: A 3 by 3 singular Markov matrix with trace $\frac{1}{2}$ has what λ 's?

- 23** Find three 2 by 2 matrices that have $\lambda_1 = \lambda_2 = 0$. The trace is zero and the determinant is zero. A might not be the zero matrix but check that $A^2 = 0$.
- 24** This matrix is singular with rank one. Find three λ 's and three eigenvectors:

$$A = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 2 & 1 & 2 \end{bmatrix}.$$

- 25** Suppose A and B have the same eigenvalues $\lambda_1, \dots, \lambda_n$ with the same independent eigenvectors $\mathbf{x}_1, \dots, \mathbf{x}_n$. Then $A = B$. Reason: Any vector $\mathbf{x}$ is a combination $c_1\mathbf{x}_1 + \dots + c_n\mathbf{x}_n$. What is $A\mathbf{x}$? What is $B\mathbf{x}$?
- 26** The block B has eigenvalues 1, 2 and C has eigenvalues 3, 4 and D has eigenvalues 5, 7. Find the eigenvalues of the 4 by 4 matrix A :

$$A = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix} = \begin{bmatrix} 0 & 1 & 3 & 0 \\ -2 & 3 & 0 & 4 \\ 0 & 0 & 6 & 1 \\ 0 & 0 & 1 & 6 \end{bmatrix}.$$

- 27** Find the rank and the four eigenvalues of A and C :

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

- 28** Subtract I from the previous A . Find the λ 's and then the determinants of

$$B = A - I = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \quad \text{and} \quad C = I - A = \begin{bmatrix} 0 & -1 & -1 & -1 \\ -1 & 0 & -1 & -1 \\ -1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 0 \end{bmatrix}.$$

- 29** (Review) Find the eigenvalues of A , B , and C :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

- 30** When $a + b = c + d$ show that $(1, 1)$ is an eigenvector and find both eigenvalues:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

- 31** If we exchange rows 1 and 2 *and* columns 1 and 2, the eigenvalues don't change. Find eigenvectors of A and B for $\lambda = 11$. *Rank one* gives $\lambda_2 = \lambda_3 = 0$.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 4 & 8 & 4 \end{bmatrix} \quad \text{and} \quad B = PAP^T = \begin{bmatrix} 6 & 3 & 3 \\ 2 & 1 & 1 \\ 8 & 4 & 4 \end{bmatrix}.$$

- 32** Suppose A has eigenvalues 0, 3, 5 with independent eigenvectors u, v, w .
- (a) Give a basis for the nullspace and a basis for the column space.
 - (b) Find a particular solution to $Ax = v + w$. Find all solutions.
 - (c) $Ax = u$ has no solution. If it did then _____ would be in the column space.
- 33** Suppose u, v are orthonormal vectors in $\mathbb{R}^2$, and $A = uv^T$. Compute $A^2 = uv^T uv^T$ to discover the eigenvalues of A . Check that the trace of A agrees with $\lambda_1 + \lambda_2$.
- 34** Find the eigenvalues of this permutation matrix P from $\det(P - \lambda I) = 0$. Which vectors are not changed by the permutation? They are eigenvectors for $\lambda = 1$. Can you find three more eigenvectors?

$$P = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

Challenge Problems

- 35** There are six 3 by 3 permutation matrices P . What numbers can be the *determinants* of P ? What numbers can be *pivots*? What numbers can be the *trace* of P ? What *four numbers* can be eigenvalues of P , as in Problem 15?
- 36** Is there a real 2 by 2 matrix (other than I) with $A^3 = I$? Its eigenvalues must satisfy $\lambda^3 = 1$. They can be $e^{2\pi i/3}$ and $e^{-2\pi i/3}$. What trace and determinant would this give? Construct a rotation matrix as A (which angle of rotation?).
- 37** (a) Find the eigenvalues and eigenvectors of A . They depend on c :

$$A = \begin{bmatrix} .4 & 1 - c \\ .6 & c \end{bmatrix}.$$

- (b) Show that A has just one line of eigenvectors when $c = 1.6$.
- (c) This is a Markov matrix when $c = .8$. Then A^n will approach what matrix A^∞ ?

6.2 Diagonalizing a Matrix

When x is an eigenvector, multiplication by A is just multiplication by a number λ : $Ax = \lambda x$. All the difficulties of matrices are swept away. Instead of an interconnected system, we can follow the eigenvectors separately. It is like having a *diagonal matrix*, with no off-diagonal interconnections. The 100th power of a diagonal matrix is easy.

The point of this section is very direct. **The matrix A turns into a diagonal matrix Λ when we use the eigenvectors properly.** This is the matrix form of our key idea. We start right off with that one essential computation.

Diagonalization Suppose the n by n matrix A has n linearly independent eigenvectors $x_1, \dots, x_n$. Put them into the columns of an **eigenvector matrix** S . Then $S^{-1}AS$ is the **eigenvalue matrix** Λ :

Eigenvector matrix S
Eigenvalue matrix Λ

$$S^{-1}AS = \Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}. \quad (1)$$

The matrix A is “diagonalized.” We use capital lambda for the eigenvalue matrix, because of the small λ ’s (the eigenvalues) on its diagonal.

Proof Multiply A times its eigenvectors, which are the columns of S . The first column of AS is Ax_1 . That is $\lambda_1 x_1$. Each column of S is multiplied by its eigenvalue λ_i :

$$A \text{ times } S \quad AS = A \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} = \begin{bmatrix} \lambda_1 x_1 & \cdots & \lambda_n x_n \end{bmatrix}.$$

The trick is to split this matrix AS into S times Λ :

$$S \text{ times } \Lambda \quad \begin{bmatrix} \lambda_1 x_1 & \cdots & \lambda_n x_n \end{bmatrix} = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = S\Lambda.$$

Keep those matrices in the right order! Then λ_1 multiplies the first column x_1 , as shown. The diagonalization is complete, and we can write $AS = S\Lambda$ in two good ways:

$$AS = S\Lambda \quad \text{is} \quad S^{-1}AS = \Lambda \quad \text{or} \quad A = S\Lambda S^{-1}. \quad (2)$$

The matrix S has an inverse, because its columns (the eigenvectors of A) were assumed to be linearly independent. *Without n independent eigenvectors, we can’t diagonalize.*

A and Λ have the same eigenvalues $\lambda_1, \dots, \lambda_n$. The eigenvectors are different. The job of the original eigenvectors $x_1, \dots, x_n$ was to diagonalize A . Those eigenvectors in S produce $A = S\Lambda S^{-1}$. You will soon see the simplicity and importance and meaning of the n th power $A^n = S\Lambda^n S^{-1}$.

Example 1 This A is triangular so the λ 's are on the diagonal: $\lambda = 1$ and $\lambda = 6$.

$$\text{Eigenvectors} \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix}$$

$S^{-1} \qquad A \qquad S \qquad \Lambda$

In other words $A = S\Lambda S^{-1}$. Then watch $A^2 = S\Lambda S^{-1}S\Lambda S^{-1}$. When you remove $S^{-1}S = I$, this becomes $S\Lambda^2 S^{-1}$. *Same eigenvectors in S and squared eigenvalues in Λ^2 .*

The k th power will be $A^k = S\Lambda^k S^{-1}$ which is easy to compute:

$$\text{Powers of } A \quad \begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix}^k = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ & 6^k \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 6^k - 1 \\ 0 & 6^k \end{bmatrix}.$$

With $k = 1$ we get A . With $k = 0$ we get $A^0 = I$ (and $\lambda^0 = 1$). With $k = -1$ we get A^{-1} . You can see how $A^2 = \begin{bmatrix} 1 & 35 \\ 0 & 36 \end{bmatrix}$ fits that formula when $k = 2$.

Here are four small remarks before we use Λ again.

Remark 1 Suppose the eigenvalues $\lambda_1, \dots, \lambda_n$ are all different. Then it is automatic that the eigenvectors $x_1, \dots, x_n$ are independent. *Any matrix that has no repeated eigenvalues can be diagonalized.*

Remark 2 *We can multiply eigenvectors by any nonzero constants.* $Ax = \lambda x$ will remain true. In Example 1, we can divide the eigenvector $(1, 1)$ by $\sqrt{2}$ to produce a unit vector.

Remark 3 The eigenvectors in S come in the same order as the eigenvalues in Λ . To reverse the order in Λ , put $(1, 1)$ before $(1, 0)$ in S :

$$\text{New order } 6, 1 \quad \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix} = \Lambda_{\text{new}}$$

To diagonalize A we *must* use an eigenvector matrix. From $S^{-1}AS = \Lambda$ we know that $AS = S\Lambda$. Suppose the first column of S is x . Then the first columns of AS and $S\Lambda$ are Ax and $\lambda_1 x$. For those to be equal, x must be an eigenvector.

Remark 4 (repeated warning for repeated eigenvalues) Some matrices have too few eigenvectors. *Those matrices cannot be diagonalized.* Here are two examples:

$$\text{Not diagonalizable} \quad A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Their eigenvalues happen to be 0 and 0. Nothing is special about $\lambda = 0$, it is the repetition of λ that counts. All eigenvectors of the first matrix are multiples of $(1, 1)$:

$$\text{Only one line of eigenvectors} \quad Ax = 0x \quad \text{means} \quad \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{and} \quad x = c \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

There is no second eigenvector, so the unusual matrix A cannot be diagonalized.

Those matrices are the best examples to test any statement about eigenvectors. In many true-false questions, non-diagonalizable matrices lead to *false*.

Remember that there is no connection between invertibility and diagonalizability:

- **Invertibility** is concerned with the *eigenvalues* ($\lambda = 0$ or $\lambda \neq 0$).
- **Diagonalizability** is concerned with the *eigenvectors* (too few or enough for S).

Each eigenvalue has at least one eigenvector! $A - \lambda I$ is singular. If $(A - \lambda I)x = \mathbf{0}$ leads you to $x = \mathbf{0}$, λ is *not* an eigenvalue. Look for a mistake in solving $\det(A - \lambda I) = 0$.

Eigenvectors for n different λ 's are independent. Then we can diagonalize A .

Independent x from different λ Eigenvectors $x_1, \dots, x_j$ that correspond to distinct (all different) eigenvalues are linearly independent. An n by n matrix that has n different eigenvalues (no repeated λ 's) must be diagonalizable.

Proof Suppose $c_1x_1 + c_2x_2 = \mathbf{0}$. Multiply by A to find $c_1\lambda_1x_1 + c_2\lambda_2x_2 = \mathbf{0}$. Multiply by λ_2 to find $c_1\lambda_2x_1 + c_2\lambda_2x_2 = \mathbf{0}$. Now subtract one from the other:

Subtraction leaves $(\lambda_1 - \lambda_2)c_1x_1 = \mathbf{0}$. Therefore $c_1 = 0$.

Since the λ 's are different and $x_1 \neq \mathbf{0}$, we are forced to this conclusion that $c_1 = 0$. Similarly $c_2 = 0$. No other combination gives $c_1x_1 + c_2x_2 = \mathbf{0}$, so the eigenvectors x_1 and x_2 must be independent.

This proof extends directly to j eigenvectors. Suppose $c_1x_1 + \dots + c_jx_j = \mathbf{0}$. Multiply by A , multiply by λ_j , and subtract. This removes x_j . Now multiply by A and by λ_{j-1} and subtract. This removes x_{j-1} . Eventually only x_1 is left:

$$(\lambda_1 - \lambda_2) \cdots (\lambda_1 - \lambda_j)c_1x_1 = \mathbf{0} \quad \text{which forces} \quad c_1 = 0. \quad (3)$$

Similarly every $c_i = 0$. When the λ 's are all different, the eigenvectors are independent. A full set of eigenvectors can go into the columns of the eigenvector matrix S .

Example 2 Powers of A The Markov matrix $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$ in the last section had $\lambda_1 = 1$ and $\lambda_2 = .5$. Here is $A = S\Lambda S^{-1}$ with those eigenvalues in the diagonal Λ :

$$\begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix} = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & .5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix} = S\Lambda S^{-1}.$$

The eigenvectors $(.6, .4)$ and $(1, -1)$ are in the columns of S . They are also the eigenvectors of A^2 . Watch how A^2 has the same S , and **the eigenvalue matrix of A^2 is Λ^2** :

Same S for A^2

$$A^2 = S\Lambda S^{-1}S\Lambda S^{-1} = S\Lambda^2 S^{-1}. \quad (4)$$

Just keep going, and you see why the high powers A^k approach a "steady state":

$$\text{Powers of } A \quad A^k = S\Lambda^k S^{-1} = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1^k & 0 \\ 0 & (.5)^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix}.$$

As k gets larger, $(.5)^k$ gets smaller. In the limit it disappears completely. That limit is A^∞ :

$$\text{Limit } k \rightarrow \infty \quad A^\infty = \begin{bmatrix} .6 & 1 \\ .4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ .4 & -.6 \end{bmatrix} = \begin{bmatrix} .6 & .6 \\ .4 & .4 \end{bmatrix}.$$

The limit has the eigenvector x_1 in both columns. We saw this A^∞ on the very first page of the chapter. Now we see it coming, from powers like $A^{100} = S\Lambda^{100}S^{-1}$.

Question When does $A^k \rightarrow$ zero matrix? **Answer** All $|\lambda| < 1$.

Fibonacci Numbers

We present a famous example, where eigenvalues tell how fast the Fibonacci numbers grow. *Every new Fibonacci number is the sum of the two previous F 's:*

The sequence 0, 1, 1, 2, 3, 5, 8, 13, ... *comes from* $F_{k+2} = F_{k+1} + F_k$.

These numbers turn up in a fantastic variety of applications. Plants and trees grow in a spiral pattern, and a pear tree has 8 growths for every 3 turns. For a willow those numbers can be 13 and 5. The champion is a sunflower of Daniel O'Connell, which had 233 seeds in 144 loops. Those are the Fibonacci numbers F_{13} and F_{12} . Our problem is more basic.

Problem: Find the Fibonacci number F_{100} The slow way is to apply the rule $F_{k+2} = F_{k+1} + F_k$ one step at a time. By adding $F_6 = 8$ to $F_7 = 13$ we reach $F_8 = 21$. Eventually we come to F_{100} . Linear algebra gives a better way.

The key is to begin with a matrix equation $\mathbf{u}_{k+1} = A\mathbf{u}_k$. That is a *one-step* rule for vectors, while Fibonacci gave a two-step rule for scalars. We match those rules by putting two Fibonacci numbers into a vector. Then you will see the matrix A .

$$\text{Let } \mathbf{u}_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}. \text{ The rule } \begin{matrix} F_{k+2} = F_{k+1} + F_k \\ F_{k+1} = F_{k+1} \end{matrix} \text{ is } \mathbf{u}_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{u}_k. \quad (5)$$

Every step multiplies by $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. After 100 steps we reach $\mathbf{u}_{100} = A^{100}\mathbf{u}_0$:

$$\mathbf{u}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{u}_3 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \dots, \quad \mathbf{u}_{100} = \begin{bmatrix} F_{101} \\ F_{100} \end{bmatrix}.$$

This problem is just right for eigenvalues. Subtract λ from the diagonal of A :

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \text{ leads to } \det(A - \lambda I) = \lambda^2 - \lambda - 1.$$

The equation $\lambda^2 - \lambda - 1 = 0$ is solved by the quadratic formula $(-b \pm \sqrt{b^2 - 4ac})/2a$:

$$\text{Eigenvalues } \lambda_1 = \frac{1 + \sqrt{5}}{2} \approx 1.618 \quad \text{and} \quad \lambda_2 = \frac{1 - \sqrt{5}}{2} \approx -.618.$$

These eigenvalues lead to eigenvectors $\mathbf{x}_1 = (\lambda_1, 1)$ and $\mathbf{x}_2 = (\lambda_2, 1)$. Step 2 finds the combination of those eigenvectors that gives $\mathbf{u}_0 = (1, 0)$:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\lambda_1 - \lambda_2} \left(\begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix} - \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix} \right) \quad \text{or} \quad \mathbf{u}_0 = \frac{\mathbf{x}_1 - \mathbf{x}_2}{\lambda_1 - \lambda_2}. \quad (6)$$

Step 3 multiplies u_0 by A^{100} to find u_{100} . The eigenvectors x_1 and x_2 stay separate! They are multiplied by $(\lambda_1)^{100}$ and $(\lambda_2)^{100}$:

$$\text{100 steps from } u_0 \quad u_{100} = \frac{(\lambda_1)^{100}x_1 - (\lambda_2)^{100}x_2}{\lambda_1 - \lambda_2}. \quad (7)$$

We want F_{100} = second component of u_{100} . The second components of x_1 and x_2 are 1. The difference between $(1 + \sqrt{5})/2$ and $(1 - \sqrt{5})/2$ is $\lambda_1 - \lambda_2 = \sqrt{5}$. We have F_{100} :

$$F_{100} = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^{100} - \left(\frac{1 - \sqrt{5}}{2} \right)^{100} \right] \approx 3.54 \cdot 10^{20}. \quad (8)$$

Is this a whole number? *Yes*. The fractions and square roots must disappear, because Fibonacci's rule $F_{k+2} = F_{k+1} + F_k$ stays with integers. The second term in (8) is less than $\frac{1}{2}$, so it must move the first term to the nearest whole number:

$$k\text{th Fibonacci number} = \frac{\lambda_1^k - \lambda_2^k}{\lambda_1 - \lambda_2} = \text{nearest integer to } \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^k. \quad (9)$$

The ratio of F_6 to F_5 is $8/5 = 1.6$. The ratio F_{101}/F_{100} must be very close to the limiting ratio $(1 + \sqrt{5})/2$. The Greeks called this number the “golden mean”. For some reason a rectangle with sides 1.618 and 1 looks especially graceful.

Matrix Powers A^k

Fibonacci's example is a typical difference equation $u_{k+1} = Au_k$. *Each step multiplies by A* . The solution is $u_k = A^k u_0$. We want to make clear how diagonalizing the matrix gives a quick way to compute A^k and find u_k in three steps.

The eigenvector matrix S produces $A = S\Lambda S^{-1}$. This is a factorization of the matrix, like $A = LU$ or $A = QR$. The new factorization is perfectly suited to computing powers, because *every time S^{-1} multiplies S we get I* :

$$\text{Powers of } A \quad A^k u_0 = (S\Lambda S^{-1}) \cdots (S\Lambda S^{-1}) u_0 = S\Lambda^k S^{-1} u_0$$

I will split $S\Lambda^k S^{-1} u_0$ into three steps that show how eigenvalues work:

1. Write u_0 as a combination $c_1 x_1 + \cdots + c_n x_n$ of the eigenvectors. Then $c = S^{-1} u_0$.
2. Multiply each eigenvector x_i by $(\lambda_i)^k$. Now we have $\Lambda^k S^{-1} u_0$.
3. Add up the pieces $c_i (\lambda_i)^k x_i$ to find the solution $u_k = A^k u_0$. This is $S\Lambda^k S^{-1} u_0$.

$$\text{Solution for } u_{k+1} = Au_k \quad u_k = A^k u_0 = c_1 (\lambda_1)^k x_1 + \cdots + c_n (\lambda_n)^k x_n. \quad (10)$$

In matrix language A^k equals $(S\Lambda S^{-1})^k$ which is S times Λ^k times S^{-1} . In Step 1,

the eigenvectors in S lead to the c 's in the combination $\mathbf{u}_0 = c_1 \mathbf{x}_1 + \cdots + c_n \mathbf{x}_n$:

$$\text{Step 1} \quad \mathbf{u}_0 = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_n \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}. \quad \text{This says that } \mathbf{u}_0 = S\mathbf{c}. \quad (11)$$

The coefficients in Step 1 are $\mathbf{c} = S^{-1}\mathbf{u}_0$. Then Step 2 multiplies by Λ^k . The final result $\mathbf{u}_k = \sum c_i (\lambda_i)^k \mathbf{x}_i$ in Step 3 is the product of S and Λ^k and $S^{-1}\mathbf{u}_0$:

$$A^k \mathbf{u}_0 = S \Lambda^k S^{-1} \mathbf{u}_0 = S \Lambda^k \mathbf{c} = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_n \end{bmatrix} \begin{bmatrix} (\lambda_1)^k & & \\ & \ddots & \\ & & (\lambda_n)^k \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}. \quad (12)$$

This result is exactly $\mathbf{u}_k = c_1 (\lambda_1)^k \mathbf{x}_1 + \cdots + c_n (\lambda_n)^k \mathbf{x}_n$. It solves $\mathbf{u}_{k+1} = A\mathbf{u}_k$.

Example 3 Start from $\mathbf{u}_0 = (1, 0)$. Compute $A^k \mathbf{u}_0$ when S and Λ contain these eigenvectors and eigenvalues:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \quad \text{has} \quad \lambda_1 = 2 \quad \text{and} \quad \mathbf{x}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \lambda_2 = -1 \quad \text{and} \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

This matrix is like Fibonacci except the rule is changed to $F_{k+2} = F_{k+1} + 2F_k$. The new numbers start 0, 1, 1, 3. They grow faster from $\lambda = 2$.

Solution in three steps Find $\mathbf{u}_0 = c_1 \mathbf{x}_1 + c_2 \mathbf{x}_2$ and then $\mathbf{u}_k = c_1 (\lambda_1)^k \mathbf{x}_1 + c_2 (\lambda_2)^k \mathbf{x}_2$

$$\text{Step 1} \quad \mathbf{u}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{so} \quad c_1 = c_2 = \frac{1}{3}$$

$$\text{Step 2} \quad \text{Multiply the two parts by } (\lambda_1)^k = 2^k \text{ and } (\lambda_2)^k = (-1)^k$$

$$\text{Step 3} \quad \text{Combine eigenvectors } c_1 (\lambda_1)^k \mathbf{x}_1 \text{ and } c_2 (\lambda_2)^k \mathbf{x}_2 \text{ into } \mathbf{u}_k:$$

$$\mathbf{u}_k = A^k \mathbf{u}_0 \quad \mathbf{u}_k = \frac{1}{3} 2^k \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{3} (-1)^k \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (13)$$

The new number is $F_k = (2^k - (-1)^k)/3$. After 0, 1, 1, 3 comes $F_4 = 15/3 = 5$.

Behind these numerical examples lies a fundamental idea: **Follow the eigenvectors**. In Section 6.3 this is the crucial link from linear algebra to differential equations (powers λ^k will become $e^{\lambda t}$). Chapter 7 sees the same idea as “transforming to an eigenvector basis.” The best example of all is a **Fourier series**, built from the eigenvectors of d/dx .

Nondiagonalizable Matrices (Optional)

Suppose λ is an eigenvalue of A . We discover that fact in two ways:

1. **Eigenvectors (geometric)** There are nonzero solutions to $Ax = \lambda x$.
2. **Eigenvalues (algebraic)** The determinant of $A - \lambda I$ is zero.

The number λ may be a simple eigenvalue or a multiple eigenvalue, and we want to know its *multiplicity*. Most eigenvalues have multiplicity $M = 1$ (simple eigenvalues). Then there is a single line of eigenvectors, and $\det(A - \lambda I)$ does not have a double factor.

For exceptional matrices, an eigenvalue can be *repeated*. Then there are two different ways to count its multiplicity. Always $\text{GM} \leq \text{AM}$ for each λ :

1. **(Geometric Multiplicity = GM)** Count the **independent eigenvectors** for λ . This is the dimension of the nullspace of $A - \lambda I$.
2. **(Algebraic Multiplicity = AM)** Count the **repetitions of λ** among the eigenvalues. Look at the n roots of $\det(A - \lambda I) = 0$.

If A has $\lambda = 4, 4, 4$, that eigenvalue has AM = 3 and GM = 1, 2, or 3.

The following matrix A is the standard example of trouble. Its eigenvalue $\lambda = 0$ is repeated. It is a double eigenvalue (AM = 2) with only one eigenvector (GM = 1).

AM = 2
GM = 1 $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has $\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ 0 & -\lambda \end{vmatrix} = \lambda^2$. **$\lambda = 0, 0$ but 1 eigenvector**

There “should” be two eigenvectors, because $\lambda^2 = 0$ has a double root. The double factor λ^2 makes $\text{AM} = 2$. But there is only one eigenvector $x = (1, 0)$. *This shortage of eigenvectors when GM is below AM means that A is not diagonalizable.*

The vector called “repeats” in the Teaching Code **eigval** gives the algebraic multiplicity AM for each eigenvalue. When repeats = [1 1 . . 1] we know that the n eigenvalues are all different and A is diagonalizable. The sum of all components in “repeats” is always n , because every n th degree equation $\det(A - \lambda I) = 0$ has n roots (counting repetitions).

The diagonal matrix $\mathbf{D}$ in the Teaching Code `eigvec` gives the geometric multiplicity GM for each eigenvalue. This counts the independent eigenvectors. The total number of independent eigenvectors might be less than n . Then A is not diagonalizable.

We emphasize again: $\lambda = 0$ makes for easy computations, but these three matrices also have the same shortage of eigenvectors. Their repeated eigenvalue is $\lambda = 5$. Traces are 10, determinants are 25:

$$A = \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 6 & -1 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 7 & 2 \\ -2 & 3 \end{bmatrix}.$$

Those all have $\det(A - \lambda I) = (\lambda - 5)^2$. The algebraic multiplicity is $\text{AM} = 2$. But each $A - 5I$ has rank $r = 1$. The geometric multiplicity is $\text{GM} = 1$. There is only one line of eigenvectors for $\lambda = 5$, and these matrices are not diagonalizable.

Eigenvalues of AB and $A + B$

The first guess about the eigenvalues of AB is not true. An eigenvalue λ of A times an eigenvalue β of B usually does *not* give an eigenvalue of AB :

False proof (14)

$$ABx = A\beta x = \beta Ax = \beta\lambda x.$$

It seems that β times λ is an eigenvalue. When x is an eigenvector for A and B , this proof is correct. *The mistake is to expect that A and B automatically share the same eigenvector x .* Usually they don't. Eigenvectors of A are not generally eigenvectors of B . A and B could have all zero eigenvalues while 1 is an eigenvalue of AB :

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}; \quad \text{then} \quad AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad A+B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

For the same reason, the eigenvalues of $A + B$ are generally not $\lambda + \beta$. Here $\lambda + \beta = 0$ while $A + B$ has eigenvalues 1 and -1 . (At least they add to zero.)

The false proof suggests what is true. Suppose x really is an eigenvector for both A and B . Then we do have $ABx = \lambda\beta x$ and $BAx = \lambda\beta x$. When all n eigenvectors are shared, we *can* multiply eigenvalues. The test $AB = BA$ for shared eigenvectors is important in quantum mechanics—time out to mention this application of linear algebra:

Commuting matrices share eigenvectors Suppose both A and B can be diagonalized. They share the same eigenvector matrix S if and only if $AB = BA$.

Heisenberg's uncertainty principle In quantum mechanics, the position matrix P and the momentum matrix Q do not commute. In fact $QP - PQ = I$ (these are infinite matrices). Then we cannot have $Px = 0$ at the same time as $Qx = 0$ (unless $x = 0$). If we knew the position exactly, we could not also know the momentum exactly. Problem 28 derives Heisenberg's uncertainty principle $\|Px\| \|Qx\| \geq \frac{1}{2}\|x\|^2$.

■ REVIEW OF THE KEY IDEAS ■

1. If A has n independent eigenvectors $x_1, \dots, x_n$, they go into the columns of S .

$$A \text{ is diagonalized by } S \quad S^{-1}AS = \Lambda \quad \text{and} \quad A = S\Lambda S^{-1}.$$

2. The powers of A are $A^k = S\Lambda^k S^{-1}$. The eigenvectors in S are unchanged.
3. The eigenvalues of A^k are $(\lambda_1)^k, \dots, (\lambda_n)^k$ in the matrix Λ^k .
4. The solution to $u_{k+1} = Au_k$ starting from u_0 is $u_k = A^k u_0 = S\Lambda^k S^{-1}u_0$:

$$u_k = c_1(\lambda_1)^k x_1 + \dots + c_n(\lambda_n)^k x_n \quad \text{provided} \quad u_0 = c_1 x_1 + \dots + c_n x_n.$$

That shows Steps 1, 2, 3 (c 's from $S^{-1}u_0$, λ^k from Λ^k , and x 's from S)

5. A is diagonalizable if every eigenvalue has enough eigenvectors (GM = AM).

■ WORKED EXAMPLES ■

6.2 A The **Lucas numbers** are like the Fibonacci numbers except they start with $L_1 = 1$ and $L_2 = 3$. Following the rule $L_{k+2} = L_{k+1} + L_k$, the next Lucas numbers are 4, 7, 11, 18. Show that the Lucas number L_{100} is $\lambda_1^{100} + \lambda_2^{100}$.

Note The key point is that $\lambda_1 + \lambda_2 = 1$ and $\lambda_1^2 + \lambda_2^2 = 3$, when the λ 's are $(1 \pm \sqrt{5})/2$. The Lucas number L_k is $\lambda_1^k + \lambda_2^k$, since this is correct for L_1 and L_2 .

Solution $u_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} u_k$ is the same as for Fibonacci, because $L_{k+2} = L_{k+1} + L_k$ is the same rule (with different starting values). The equation becomes a 2 by 2 system:

$$\text{Let } u_k = \begin{bmatrix} L_{k+1} \\ L_k \end{bmatrix}. \quad \text{The rule } \begin{array}{l} L_{k+2} = L_{k+1} + L_k \\ L_{k+1} = L_{k+1} \end{array} \quad \text{is } u_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} u_k.$$

The eigenvalues and eigenvectors of $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ still come from $\lambda^2 = \lambda + 1$:

$$\lambda_1 = \frac{1 + \sqrt{5}}{2} \quad \text{and} \quad x_1 = \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix} \qquad \lambda_2 = \frac{1 - \sqrt{5}}{2} \quad \text{and} \quad x_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}.$$

Now solve $c_1 x_1 + c_2 x_2 = u_1 = (3, 1)$. The solution is $c_1 = \lambda_1$ and $c_2 = \lambda_2$. Check:

$$\lambda_1 x_1 + \lambda_2 x_2 = \begin{bmatrix} \lambda_1^2 + \lambda_2^2 \\ \lambda_1 + \lambda_2 \end{bmatrix} = \begin{bmatrix} \text{trace of } A^2 \\ \text{trace of } A \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} = u_1$$

$u_{100} = A^{99} u_1$ tells us the Lucas numbers (L_{101}, L_{100}) . The second components of the eigenvectors x_1 and x_2 are 1, so the second component of u_{100} is the answer we want:

Lucas number $L_{100} = c_1 \lambda_1^{99} + c_2 \lambda_2^{99} = \lambda_1^{100} + \lambda_2^{100}.$

Lucas starts faster than Fibonacci, and ends up larger by a factor near $\sqrt{5}$.

6.2 B Find the inverse and the eigenvalues and the determinant of A :

$$A = 5 * \text{eye}(4) - \text{ones}(4) = \begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}.$$

Describe an eigenvector matrix S that gives $S^{-1}AS = \Lambda$.

Solution What are the eigenvalues of the all-ones matrix **ones**(4)? Its rank is certainly 1, so three eigenvalues are $\lambda = 0, 0, 0$. Its trace is 4, so the other eigenvalue is $\lambda = 4$. Subtract this all-ones matrix from $5I$ to get our matrix A :

Subtract the eigenvalues 4, 0, 0, 0 from 5, 5, 5, 5. The eigenvalues of A are 1, 5, 5, 5.

The determinant of A is 125, the product of those four eigenvalues. The eigenvector for $\lambda = 1$ is $\mathbf{x} = (1, 1, 1, 1)$ or (c, c, c, c) . The other eigenvectors are perpendicular to $\mathbf{x}$ (since A is symmetric). The nicest eigenvector matrix S is the symmetric orthogonal Hadamard matrix H (normalized to unit column vectors):

$$\text{Orthonormal eigenvectors } S = H = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} = H^T = H^{-1}.$$

The eigenvalues of A^{-1} are $1, \frac{1}{5}, \frac{1}{5}, \frac{1}{5}$. The eigenvectors are not changed so $A^{-1} = H\Lambda^{-1}H^{-1}$. The inverse matrix is surprisingly neat:

$$A^{-1} = \frac{1}{5} * (\text{eye}(4) + \text{ones}(4)) = \frac{1}{5} \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

A is a rank-one change from $5I$. So A^{-1} is a rank-one change $I/5 + \text{ones}/5$.

The determinant 125 counts the “spanning trees” in a graph with 5 nodes (all edges included). *Trees have no loops* (graphs and trees are in Section 8.2).

With 6 nodes, the matrix $6 * \text{eye}(5) - \text{ones}(5)$ has the five eigenvalues 1, 6, 6, 6, 6.

Problem Set 6.2

Questions 1–7 are about the eigenvalue and eigenvector matrices Λ and S .

- 1 (a) Factor these two matrices into $A = S\Lambda S^{-1}$:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}.$$

(b) If $A = S\Lambda S^{-1}$ then $A^3 = () () ()$ and $A^{-1} = () () ()$.

- 2 If A has $\lambda_1 = 2$ with eigenvector $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\lambda_2 = 5$ with $\mathbf{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, use $S\Lambda S^{-1}$ to find A . No other matrix has the same λ 's and $\mathbf{x}$'s.
- 3 Suppose $A = S\Lambda S^{-1}$. What is the eigenvalue matrix for $A + 2I$? What is the eigenvector matrix? Check that $A + 2I = () () ()^{-1}$.

- 4 True or false: If the columns of S (eigenvectors of A) are linearly independent, then
- (a) A is invertible (b) A is diagonalizable
 (c) S is invertible (d) S is diagonalizable.
- 5 If the eigenvectors of A are the columns of I , then A is a _____ matrix. If the eigenvector matrix S is triangular, then S^{-1} is triangular. Prove that A is also triangular.
- 6 Describe all matrices S that diagonalize this matrix A (find all eigenvectors):

$$A = \begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix}.$$

Then describe all matrices that diagonalize A^{-1} .

- 7 Write down the most general matrix that has eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Questions 8–10 are about Fibonacci and Gibonacci numbers.

- 8 Diagonalize the Fibonacci matrix by completing S^{-1} :

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} & \\ & \end{bmatrix}.$$

Do the multiplication $S\Lambda^k S^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ to find its second component. This is the k th Fibonacci number $F_k = (\lambda_1^k - \lambda_2^k) / (\lambda_1 - \lambda_2)$.

- 9 Suppose G_{k+2} is the *average* of the two previous numbers G_{k+1} and G_k :

$$\begin{aligned} G_{k+2} &= \frac{1}{2}G_{k+1} + \frac{1}{2}G_k \\ G_{k+1} &= G_{k+1} \end{aligned} \quad \text{is} \quad \begin{bmatrix} G_{k+2} \\ G_{k+1} \end{bmatrix} = \begin{bmatrix} & \\ A & \end{bmatrix} \begin{bmatrix} G_{k+1} \\ G_k \end{bmatrix}.$$

- (a) Find the eigenvalues and eigenvectors of A .
 (b) Find the limit as $n \rightarrow \infty$ of the matrices $A^n = S\Lambda^n S^{-1}$.
 (c) If $G_0 = 0$ and $G_1 = 1$ show that the Gibonacci numbers approach $\frac{2}{3}$.
- 10 Prove that every third Fibonacci number in $0, 1, 1, 2, 3, \dots$ is even.

Questions 11–14 are about diagonalizability.

- 11 True or false: If the eigenvalues of A are $2, 2, 5$ then the matrix is certainly
- (a) invertible (b) diagonalizable (c) not diagonalizable.
- 12 True or false: If the only eigenvectors of A are multiples of $(1, 4)$ then A has
- (a) no inverse (b) a repeated eigenvalue (c) no diagonalization $S\Lambda S^{-1}$.

- 13** Complete these matrices so that $\det A = 25$. Then check that $\lambda = 5$ is repeated—the trace is 10 so the determinant of $A - \lambda I$ is $(\lambda - 5)^2$. Find an eigenvector with $Ax = 5x$. These matrices will not be diagonalizable because there is no second line of eigenvectors.

$$A = \begin{bmatrix} 8 & \\ & 2 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 9 & 4 \\ & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 10 & 5 \\ -5 & \end{bmatrix}$$

- 14** The matrix $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ is not diagonalizable because the rank of $A - 3I$ is _____. Change one entry to make A diagonalizable. Which entries could you change?

Questions 15–19 are about powers of matrices.

- 15** $A^k = S\Lambda^k S^{-1}$ approaches the zero matrix as $k \rightarrow \infty$ if and only if every λ has absolute value less than _____. Which of these matrices has $A^k \rightarrow 0$?

$$A_1 = \begin{bmatrix} .6 & .9 \\ .4 & .1 \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} .6 & .9 \\ .1 & .6 \end{bmatrix}.$$

- 16** (Recommended) Find Λ and S to diagonalize A_1 in Problem 15. What is the limit of Λ^k as $k \rightarrow \infty$? What is the limit of $S\Lambda^k S^{-1}$? In the columns of this limiting matrix you see the _____.

- 17** Find Λ and S to diagonalize A_2 in Problem 15. What is $(A_2)^{10}u_0$ for these u_0 ?

$$u_0 = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \text{and} \quad u_0 = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \quad \text{and} \quad u_0 = \begin{bmatrix} 6 \\ 0 \end{bmatrix}.$$

- 18** Diagonalize A and compute $S\Lambda^k S^{-1}$ to prove this formula for A^k :

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \quad \text{has} \quad A^k = \frac{1}{2} \begin{bmatrix} 1 + 3^k & 1 - 3^k \\ 1 - 3^k & 1 + 3^k \end{bmatrix}.$$

- 19** Diagonalize B and compute $S\Lambda^k S^{-1}$ to prove this formula for B^k :

$$B = \begin{bmatrix} 5 & 1 \\ 0 & 4 \end{bmatrix} \quad \text{has} \quad B^k = \begin{bmatrix} 5^k & 5^k - 4^k \\ 0 & 4^k \end{bmatrix}.$$

- 20** Suppose $A = S\Lambda S^{-1}$. Take determinants to prove $\det A = \det \Lambda = \lambda_1 \lambda_2 \cdots \lambda_n$. This quick proof only works when A can be _____.

- 21** Show that $\text{trace } ST = \text{trace } TS$, by adding the diagonal entries of ST and TS :

$$S = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad T = \begin{bmatrix} q & r \\ s & t \end{bmatrix}.$$

Choose T as ΛS^{-1} . Then $S\Lambda S^{-1}$ has the same trace as $\Lambda S^{-1}S = \Lambda$. The trace of A equals the trace of Λ = sum of the eigenvalues.